A certified resource ledger shows where ZX rewriting pays on Quantinuum H2

Team Learned ZX Rewriting. Quantinuum Singapore Grand Challenge, submission 1, draft v5 (8 October 2026).

Tags: [F] marks a statement verifiable from an artifact in the repository or from the cited source. [O] marks a judgement or a plan. Every number in a results table of Section 3 is produced by `report/make_tables.py` from `results/*.csv`. The emulator runs are reported in Sections 3.7 and 3.7.1.

Scoring map

[F] The four criteria and their weights are those of the track rubric, read from the Aqora GraphQL API on 8 October 2026 and stored in `docs/rubric.md`. [O] The table assigns the evidence for each criterion to a section of this report.

Criterion	Weight	Where the evidence is
Problem & Value	30 %	Section 1, Section 6
Technical Performance & Hardware Use	30 %	Section 3 (count tables A1 to B2 and the count-only table B1x, proxy tables A4 and A5, Selene tables S1 and S2, H2-Emulator table E1, H2-2E tables E2 and E3), Section 2.6 and `results/selene.md` for the Selene runs, Sections 3.7 and 3.7.1 and `results/emulator_jobs.csv` for the platform jobs, notebook `report/notebook_skeleton.py`
Scientific Merit	20 %	Section 2.5 (certificate), Section 3.4 and `results/certificates.md`, arXiv:2608.22801 and arXiv:2608.22810
Engineering & Reproducibility	20 %	Section 5 (environment, seeds, script table, test table, verification reports), `README.md`, the tests under `bench/`, `noise/`, `barrier/`, `pipelines/`, `selene/`, `model/`, `submit/` and `value/`, Section 4.7 for the refit command

1. Problem and value

[F] The HQC, shot and T-count numbers below come from the script outputs `value/value_ledger.md` and `value/value_by_application.md`. The noise parameters come from `docs/platform_notes.md` section 3.3. Manuscript statements are cited by their labels in arXiv:2608.22801. P0 is pytket's Quantinuum pass at level 2. P2 runs PyZX teleport_reduce before that pass. HQC figures are for 1000 shots.

- [O] User. A team that runs Clifford+T subroutines on Quantinuum H2 under a fixed HQC budget and will later price them on a fault-tolerant Helios.
- [O] Decision. Should this circuit go through ZX rewriting before pytket's Quantinuum pass?
- [F] H2 cost. On the 15 checked B1 circuits P2 costs 14,772.8 HQC against 15,516.0 for P0. The saving is 743.2 HQC or 4.8 %. The median per-circuit saving is 3.9 %.
- [F] Fault-tolerant cost. ZX removes 2,862 of 7,048 T gates on B1. Each T gate is one magic state, so the saving is 40.6 %. On the four B1 circuits with a certificate the certified fraction of the remaining T-count is 0.443. The fraction 0.997 holds for the 20 synthesised B2 Haar circuits at n = 2 and 3 and not for the B1 circuits.
- [F] Algorithms. Every circuit except hwb6, hwb8 and the two ham15 circuits is an adder, multiplier, Fourier transform, oracle, search instance or multi-controlled Toffoli. The adders, multipliers and the Fourier transform come from factoring and discrete-log circuits. grover_5 and mod5_4 come from search (Section 1.4). [O] The Toffoli circuits are placed in search as oracle and diffusion parts.

1.1 The user and the decision

[O] The user pays each H2 run in HQC from a fixed budget. On a fault-tolerant machine the same subroutines are priced in T gates. [F] The track budget is 10,000 HQC per team on H2-1E and H2-2E according to the Aqora forum topic "Emulator Cost" by Irfan Khan of Quantinuum (Section 3.7). No copy is in the repository.

[O] The decision is made per circuit: run P0 alone or insert the ZX stage first. The ledger answers it in HQC and in magic states. It also says when the answer is no.

[F] No user interview or survey is part of this submission. [O] The need rests on the H2 cost formula and on the algorithm literature of Section 1.4. None of the six papers of Section 3.4 compares ZX rewriting against pytket's Quantinuum pass on the native H2 gate set.

1.2 What the pipeline saves on H2 today

[F] The organisers' sample code prices an H2 program as HQC = 5 + N_shots (N_1q + 10 N_2q + 5 N_M) / 5000. N_1q counts PhasedX gates. N_2q counts ZZPhase and ZZMax gates. N_M counts initialisations, measurements and resets. Rz gates are free. The source is `estimate_cost` in `src/python/pytket_cost_estimation.py` of `Quantinuum/sg_grand_challenge`. `pipelines/compile.py:hqc` reproduces that function. [O] We take this as the H2 price. [F] The Helios costing page gives the same form with maxima over 10 shots in its equation (1). See Reference 6 and `docs/platform_notes.md` section 3.4.

[F] p2 = 1.29e-3 is the largest gate fault rate of the H2-2E emulator. The H2-2E emulator parameters are p2 = 1.29e-3 and p1 = 7.3e-5 per gate, readout flips 9e-4 and 1.8e-3 for states 0 and 1 and p_init = 4e-5 (`docs/platform_notes.md` section 3.3).

[F] Table 1.1. HQC at 1000 shots by application area. B1 is checked. B1x is count-only and is never merged with B1. Saved % is computed on the area sums. Source: `value/value_by_application.md`.

Area	Family	Circuits	HQC P0	HQC P2	HQC saved	Saved %	Median per-circuit %
Integer factoring (Shor)	B1	4	592.2	545.6	46.6	7.9	4.2
Elliptic-curve discrete log, GF(2^m)	B1	1	216.2	215.2	1.0	0.5	0.5
Unstructured search (Grover)	B1	8	1,183.6	1,127.0	56.6	4.8	3.0

Reversible logic, no sourced algorithm	B1	2	13,524.0	12,885.0	639.0	4.7	4.5
All areas	B1	15	15,516.0	14,772.8	743.2	4.8	3.9
Integer factoring (Shor)	B1x	2	386.2	380.2	6.0	1.6	1.4
Elliptic-curve discrete log, GF(2^m)	B1x	2	783.0	776.4	6.6	0.8	0.9
Unstructured search (Grover)	B1x	2	632.0	610.0	22.0	3.5	3.7
Reversible logic, no sourced algorithm	B1x	2	1,487.6	1,374.4	113.2	7.6	6.2
All areas	B1x	8	3,288.8	3,141.0	147.8	4.5	2.4

[F] P2 is cheaper than P0 in HQC on 14 of the 15 B1 circuits. hwb8 alone accounts for 84.6 % of the B1 saving but serves no sourced quantum algorithm. Without hwb8 the B1 saving is 114.6 HQC. [O] The median of 3.9 % is the better guide for a single circuit.

[F] Saved HQC buys shots. The following figures assume one job that pays the 5 HQC job fee once. At the 10,000 HQC team budget P0 affords 68,178 shots of vbe_adder_3 and P2 affords 87,675. That is 19,497 extra shots or 28.6 % more. vbe_adder_3 has the largest per-circuit percentage saving of the 15 B1 circuits. The median extra-shot gain over the 15 circuits is 4.3 % (`value/value_by_application.md`). On the Grover oracle mod5_4 the gain is 11,757 shots or 7.5 %.

[F] The H2 hardware guide limits a job to 10,000 shots (`docs/platform_notes.md` section 3.4). With that limit and one fee per job P0 affords 67,974 shots of vbe_adder_3 and P2 affords 87,324. That is 19,350 extra shots or 28.5 % more. On mod5_4 the gain is 11,584 shots or 7.4 % (`value/value_by_application.md`). Whether the per-job limit applies on H2-2E is Unverifiable from the repository.

[F] The ledger also says when not to run ZX. On the 180 Trotter circuits of family A P2 costs 17,453.0 HQC against 16,422.6 for P0. That is 1,030.4 HQC or 6.3 % more. P2 is dearer on 108 circuits and cheaper on 57. On the 60 XXZ circuits P2 is dearer on every circuit and the total rises by 12.3 %. [O] A user with a Trotter circuit should skip the ZX stage.

1.3 What it saves on the fault-tolerant layer

[F] In a surface-code machine each T gate consumes one magic state (Litinski, Quantum 3, 128 (2019), arXiv:1808.02892). Magic-state counts below therefore equal T-counts. On H2 a T gate is a free Rz and changes no H2 cost.

[F] Table 1.2. T-count written and after the ZX stage of P2, by application area. Source: `value/value_by_application.md`.

Area	Family	T written	T after ZX	Magic states saved	Saved %
Integer factoring (Shor)	B1	307	199	108	35.2
Elliptic-curve discrete log, GF(2^m)	B1	112	68	44	39.3
Unstructured search (Grover)	B1	637	327	310	48.7
Reversible logic, no sourced algorithm	B1	5,992	3,592	2,400	40.1
All areas	B1	7,048	4,186	2,862	40.6
Integer factoring (Shor)	B1x	147	109	38	25.9
Elliptic-curve discrete log, GF(2^m)	B1x	427	265	162	37.9
Unstructured search (Grover)	B1x	343	171	172	50.1
Reversible logic, no sourced algorithm	B1x	735	309	426	58.0
All areas	B1x	1,652	854	798	48.3

[F] The exactness-barrier manuscript gives a per-instance floor (Kam et al., arXiv:2608.22801). The least denominator exponent of the exact Pauli channel of a Clifford+T circuit over Z[sqrt 2] is a lower bound on the T-count of every semantics-exact rewrite. The statements are Theorem B, Corollary cert and Theorem C of arXiv:2608.22801.

[F] On the 20 synthesised Haar circuits of family B2 the ZX stage removes 862 of 42,396 T gates or 2.0 %. The certified floor is 41,414 against 41,534 T gates after ZX. The certified-incompressible fraction is 0.997. Further exact T optimisation of these instances can gain at most 0.3 % of the remaining T-count. On the worst single instance the bound is 0.7 %.

[F] On the four certified B1 circuits the floor is 47 against 106 T gates after ZX. The fraction is 0.443. [O] The certificate does not say whether this gap is headroom for a better rewrite or slack in the bound (Section 3.4).

[F] Converting magic states into Helios time or cost needs factory and logical-cycle figures that Quantinuum has not published in any source we found. No conversion is made. As context only, Dasu et al. report a logical magic state on H1-1 with infidelity 7(+3/-1) x 10^-5 (arXiv:2506.14688).

1.4 Where these circuits sit in real algorithms

[F] Table 1.3. All B1 and B1x circuit names appear in Table II of Amy and Mosca (arXiv:1601.07363). Their Grover_5 is a different circuit (Section 3.4). Each circuit is counted in the first area of the summary table of `docs/applications.md` that lists it.

Area	Our circuits	Role	Citation	Status
Integer factoring (Shor)	vbe_adder_3, mod_mult_55, mod_red_21, qft_4 (B1). rc_adder_6, csla_mux_3 (B1x)	Adders and modular arithmetic of modular exponentiation. Fourier transform	arXiv:quant-ph/9511018, arXiv:1202.6614, arXiv:1710.07345, arXiv:quant-ph/0201067	[F]
Elliptic-curve discrete log, GF(2^m)	gf2_pow_4_mult (B1). gf2_pow_5_mult, gf2_pow_6_mult (B1x)	Binary-field multiplier	arXiv:0710.1093	[F]
Unstructured search (Grover)	grover_5, mod5_4 (B1)	Full search instance. Divisibility-by-5 oracle	arXiv:quant-ph/9605043, arXiv:1601.07363, reversiblebenchmarks.github.io/mod5.html	[F]
Unstructured search (Grover)	tof_3..5, barenco_tof_3..5 (B1). tof_10, barenco_tof_10 (B1x)	Multi-controlled Toffoli in oracle and diffusion	arXiv:1703.10535, arXiv:quant-ph/9503016	[O] placement
Reversible logic, no sourced algorithm	hwb6, hwb8 (B1). ham15-low, ham15-med (B1x)	Reversible-logic stress tests	reversiblebenchmarks.github.io hwbs and ham15 pages	[F] no algorithm role found

[O] The three adders also fit arithmetic oracles for chemistry and simulation (arXiv:1805.12445, arXiv:1807.03206). They are counted

once under factoring.

[F] Unverifiable items. The function computed by mod_red_21 beyond its name is Unverifiable. Whether arXiv:0710.1093 is the exact work Amy, Maslov and Mosca cite for the field multipliers is Unverifiable.

1.5 Value statement and scope of claims

[0] For an H2 user the ledger is a reproducible answer to "should I run ZX before pytket" in HQC per area. [F] P2 saves 7.9 % on the checked factoring circuits and 4.8 % on the checked search circuits. [0] On Trotter circuits the answer is no. [F] The ledger gives magic-state counts for every B1, B1x and B2 circuit. The certificate covers only the 24 circuits of `results/certificates.csv`, all with n at most 5. Four of them are B1 subroutines with a pooled certified fraction of 0.443. The other 20 are the synthesised B2 Haar circuits at n = 2 and 3 with a pooled fraction of 0.997.

[F] The winner of the finals receives credits on the on-premise Helios machine in Singapore according to the organisers' timeline email of 25 Aug 2026. No copy is in the repository. [0] Quantinuum's Helios roadmap targets fault tolerance. The T-count column prices a later fault-tolerant layer. [F] The Helios HQC formula charges no Rz gate (`docs/platform_notes.md` section 3.4). The T-count therefore does not change the cost of the finals credits.

[F] Model prediction. A proxy fit of 1.18e-3 per ZZ gate predicts a relative fidelity gain of 1.79 % for P2 on vbe_adder_3 and a median relative gain of 0.24 % over B1. [0] These are model predictions and not measurements.

[F] The 18 H2-2E jobs carry HQC estimates that sum to 5933.2 HQC. Six ran vbe_adder_3, hwb6 and grover_5 under P0 and P2 at 1000 shots. Twelve repeated vbe_adder_3 and grover_5 three more times. At 1000 shots the emulator does not resolve P2 against P0 on any of the three circuits. At 4000 shots per pipeline the pooled F_H of vbe_adder_3 is 0.9520 under P0 and 0.9628 under P2. The difference of +0.0108 has z = +2.38 and a Holm-adjusted p of 0.035 over the two pooled circuits. This measured relative gain of 1.13 % lies below the predicted 1.79 % by 1.38 sd of the difference. On grover_5 the pooled difference is +0.0007 with z = +0.10 and is not resolved at 4000 shots per pipeline. It lies 2.94 sd of the difference below the predicted gain of 0.0229 (Section 3.7.1).

[F] The ledger states value in HQC, shots and magic states and makes no claim in currency. [0] The team name Learned ZX Rewriting names the research programme. [F] The learned component of this submission is the cost model of Section 4, which predicts fidelity from compile-time features. This submission contains no learned rewriting.

2. Method

2.1 Circuit families

[F] Family A consists of 180 Trotter spin-chain circuits generated by `bench/gen_trotter.py`. Three models: the transverse-field Ising chain with open boundary, the XXZ chain started from the Neel state, and the transverse-field Ising ladder. Sizes n in {4, 6, 8, 10, 12}. Trotter steps r in {1, 2, 4} with dt = 0.1. Couplings are uniform (one instance per group) or drawn from [0.5, 1.5] with `numpy.random.default_rng(20261006)` (three instances per group). Every two-body Pauli exponential is written as a cx-rz-cx block with basis changes. The circuit as written therefore has n_cx = 2 x (number of ZZ-type terms). Ideal output distributions for n <= 10 are stored as `.npy` references.

[F] Family B1, 15 standard Clifford+T benchmark circuits with n <= 12 from the Quantinuum pyzx fork at commit e695d6d7 (`third_party/pyzx-fork/circuits/qasm`), built by `bench/build_family_B1.py`. Every Toffoli is expanded into the 7-T decomposition by pyzx so that the file as written is Clifford+T. The manifest `bench/family_B1/manifest.csv` records the source file, its commit and the counts before and after expansion, for example a T-count as written of 21 for tof_3 and 28 for mod5_4.

[F] Family B1x consists of 8 further Clifford+T benchmark circuits from the same pyzx fork at commit e695d6d7, selected by `bench/build_family_B1x.py` with the window 13 <= n <= 20 and at most 3000 gates after Toffoli expansion. The circuits are rc_adder_6 (n = 14), csla_mux_3 (15), gf2_pow_5_mult (15), ham15-low (17), ham15-med (17), gf2_pow_6_mult (18), barenco_tof_10 (19) and tof_10 (19). The qubit numbers reached run from 14 to 19 and the largest circuit is ham15-med with 1600 gates after expansion. The conversion and the manifest columns are those of family B1 (`bench/family_B1x/manifest.csv`). These circuits are compiled count-only under P0ref, P0 and P2 with the `--no-check` flag of `pipelines/run_bench_B.py`. No unitary, statevector or distribution check was run on them at this size and only the gate-set check applies. Every row of `results/family_B1x_counts.csv` therefore carries the status `ok-unchecked` and the family is reported separately from the checked families in Section 3.2.1.

[F] Family B2 consists of 20 Clifford+T circuits synthesised from Haar-random unitaries by `bench/gen_haar_b2.py`. There are five targets at n = 2 and five at n = 3 drawn with the same seed. Each target is synthesised at eps = 1e-3 and at eps = 1e-5. The construction is qiskit `qs_decomposition` followed by `transpile` to the basis {rz, sx, x, cx} with `seed_transpiler=0`. Every rz angle that is not a multiple of pi/4 is then replaced by a pygridsynth (Ross-Selinger) approximation to precision eps. The operator-norm error of each written circuit against its target is stored in the manifest, with the triangle-inequality bound n_rz x eps. This family reproduces the construction of Table E6 in the barrier manuscript.

2.2 Pipelines

[F] All pipelines are in `pipelines/compile.py` and target the H2 gate set {Rz, PhasedX, ZZPhase, ZZMax, Measure, Reset} with all-to-all connectivity. Seven pipelines run on families A and B1, five (P0ref, P0, P1, P2, P3) on family B2 and three (P0ref, P0, P2) on the count-only family B1x.

Name	Stages
P0ref	DecomposeBoxes, AutoRebase to the native set, SquashRzPhasedX, RemoveRedundancies. The circuit as written, in native gates, with no optimisation.
P0	pytket-quantinuum 0.59.3 `default_compilation_pass(optimisation_level=2)`, reproduced pass by pass from the installed source without a backend object (FullPeepholeOptimise to TK2, NormaliseTK2, DecomposeTK2 to ZZPhase, AutoRebase, squash, RemoveRedundancies). P0 is the baseline.
P0L1	The same at optimisation level 1.
P1	PyZX `full_reduce`, `extract_circuit`, `basic_optimization`, then P0.
P2	PyZX `teleport_reduce`, `Circuit.from_graph`, `basic_optimization`, then P0.

P3	pytket `ZXGraphlikeOptimisation`, then P0.
P4	The ZX stage of P1, then P0 without FullPeepholeOptimise.

[F] The ZX stages run on the circuit as written, in the gate set the ZX tools accept, and the pytket pass runs afterwards on the result. Angles are passed to PyZX through QASM, and the rounding introduced by PyZX's float-to-fraction conversion is recorded per circuit in the column `pyzx_angle_err` of `results/family_A_counts.csv`. Its maximum over the 1260 family A rows is 1.48e-10 rad, which is 4.7e-11 x pi.

2.3 Metrics

[F] For every compiled circuit `pipelines/compile.py:metrics` records n_2q (ZZPhase plus ZZMax count), the two-qubit depth, the PhasedX and Rz counts and the total gate count and depth. It also records the HQC at N_shots = 1000 with N_M = 2 n_qubits, one initialisation and one terminal measurement per qubit. The T-count after the ZX stage is counted on the PyZX circuit before any pytket pass, as the number of phase gates with angle an odd multiple of pi/4. After rebasing to the H2 gate set the T-count is undefined and is not reported.

[F] The fidelity metrics are the Hellinger fidelity F_H = (sum_i sqrt(p_i q_i))^2 and the total variation distance TVD = 0.5 sum_i |p_i - q_i| between a noisy or measured distribution q and the ideal distribution p from the `.npy` references. Both are reported for the local density-matrix proxy and for the Selene runs of Section 2.6.

[F] Both metrics are reported for the H2-Emulator output in Section 3.7. F_H is reported for the H2-2E output in Section 3.7.1, and TVD is in `results/emulator_nexus_H2-2E.csv`.

2.4 Exactness checks

[F] Every compiled circuit is checked against the P0ref compilation of the same input. The check is the unitary deviation 1 - |tr(U^dagger V)| / 2^n for n <= 8 and the statevector infidelity on |0...0> beyond, with tolerance 1e-8. Where a `.npy` reference exists, the maximum difference between the compiled output distribution and that reference is recorded as well. The measured QASM written for submission is re-simulated independently (`qasm_dist_maxdiff`). All 1260 family A rows, all 100 family B2 rows and all 105 family B1 rows pass. The two hwb8 rows under P1 and P2 come from the 5400 s run of Section 5 and have `check_dev` at most 1.7e-13. The 24 rows of family B1x were compiled with `--no-check` and carry no such check (Section 3.2.1).

[F] Family B2 is additionally checked against the Haar target it approximates. The largest deviation 1 - |tr(U_target^dagger U_compiled)| / 2^n over all 100 rows is 7.1e-6, within the gridsynth bound.

2.5 The certificate

[F] The barrier manuscript (arXiv:2608.22801) gives the floor in Theorem B(a) for one qubit and in Corollary cert for any n. Theorem C(c) gives the removable-fraction bound. Let C be an n-qubit Clifford+T circuit and let O be any optimiser that preserves its unitary exactly. Then the T-count of O(C) is at least lde(C^). Here C^ is the exact 4^n x 4^n Pauli-channel matrix of C with entries in Z[sqrt 2]. lde(C^) is the least e such that C^ = (P + Q sqrt 2) / sqrt 2^e with integer matrices P and Q. The fraction of T-count that any exact optimiser can remove is at most 1 - lde(C^) / t(C). The floor is computed from any word realising C, in one exact channel multiplication per gate, and does not depend on the word chosen.

[F] `barrier/certificate.py` implements the floor in exact integer arithmetic, with the single-qubit linkage procedure and the residue automaton of the manuscript as cross-checks. `barrier/test_certificate.py` reproduces the manuscript's finite verifications. These are the Clifford group order 24, the residue automaton with 24 states and 1536 transitions and the 18384 Matsumoto-Amano words with t <= 8. `barrier/NOTES.md` maps every implemented quantity to the line of the manuscript it comes from. These line numbers refer to the team's local TeX source, which is not distributed.

[F] The certificate was computed for the 20 family B2 circuits and for the four family B1 circuits at n = 5, where the 4^n x 4^n channel is small. The output per circuit is the floor, the T-count after the ZX stage of P1 and of P2 and the ratio floor / T_zx. The 24 rows are in `results/certificates.csv` and `results/certificates.md`. The invariant floor <= T_zx holds on all 24 rows, and P1 and P2 give the same T_zx on every row. Section 3.4 reports the rows.

[F] The manuscript is not re-derived here, with one exception that needs stating. While implementing the per-step bookkeeping of Lemma mqval(b), the implementation found that the "reduction fires exactly once" half of that lemma fails on the word T T (`barrier/NOTES.md`, section 8). The floor of Corollary cert does not use that half, and the manuscript's own text says nothing downstream depends on it. [O] The correction is being made in the manuscript.

2.6 Hardware use

[F] Two emulator tiers exist (`docs/platform_notes.md`, section 3.1). `H2-Emulator` is hosted by Nexus and costed in seconds. `H2-2E` is a hardware-tier emulator costed in HQCs with the published H2-2E noise parameters. Quantinuum documents that the syntax checker `H2-2SC` returns the HQC cost of a circuit without charge. Our one H2-2SC job returned no cost value (Section 3.7).

[F] Quantinuum's local emulator Selene (`selene-sim` 0.3.4 with `guppylang` 1.1.2) ran 468 noiseless runs, 68 depolarising runs in a first comparison and 468 depolarising runs on the full set on the team's container. The route reads `results/compiled/<pipeline>/<file>.qasm` with `pytket.qasm.circuit_from_qasm` and loads the circuit into a guppy program with `guppylang.guppy.load_pytket`. The program is built by `selene_sim.build` and run by `run_shots` on the Quest statevector backend at 2000 shots. `load_pytket` accepted the PhasedX, Rz, ZZPhase and Measure gates of every compiled circuit without rebase. The noiseless runs cover 144 family A circuits with n <= 10 and 12 family B1 circuits with n <= 10 under P0, P1 and P2. No run was flagged by the criterion TVD > 4 x sqrt(support / (2 pi N_shots)) and no shot landed outside the exact support. Scripts are in `selene/`, data in `results/selene_noiseless.csv` and `results/selene_depolarizing.csv`, and the summary in `results/selene.md`.

[F] The 68 depolarising runs use Selene's `DepolarizingErrorModel` at the H2-2E rates p_1q = 7.3e-5 and p_2q = 1.29e-3 on 17 circuits under P0 and P2, in two variants of 34 runs. The variant `gates_meas_init` adds p_meas = 1.35e-3 and p_init = 4e-5, where 1.35e-3 is the mean of the two H2-2E readout rates because Selene takes one symmetric rate. In that variant 32 of 34 runs agree with the local

density-matrix proxy within two sampling standard deviations. The variant `gates_only` leaves readout and initialisation noiseless and is not the like-for-like comparison.

[F] The full depolarising run has 468 runs on the same 156 circuits under P0, P1 and P2 with the `gates_meas_init` rates at 2000 shots. Section 3.6 and Table S2 report it.

[F] The local proxy `noise/dm_sim.py` is a density-matrix simulator. It applies a depolarising channel after every PhasedX and ZZPhase gate and adds an asymmetric readout flip and an initialisation flip. The parameters are the published H2-2E values p1, p2, p_meas and p_init from `noise/h2_params.py:H2_2E_EMULATOR`. The emission, crosstalk, dephasing and angle-scaling entries of the H2-2E model are not modelled. `noise/run_proxy.py` produced `results/family_A_proxy_h22e.csv` (540 rows, 108 circuits x 5 pipelines) and `results/family_B1_proxy_h22e.csv` (35 rows, 7 circuits x 5 pipelines). The two files feed Tables A4 and A5 and the cost model of Section 4.

[F] Platform jobs started on 8 October 2026 through `aqora` 0.35.1 as team `learned_zx_rewriting`. The scripts `submit/syntax_check.py` and `submit/run_emulator.py` submit the compiled and measured QASM files in `results/compiled/<pipeline>/` at 1000 shots with `options={"noisy_simulation": True}`. One H2-2SC syntax check and three H2-Emulator jobs with 22 circuit rows completed. Section 3.7 and Table E1 report them. Eighteen H2-2E jobs completed with an HQC estimate of 5933.2 in total. Six are the original runs of Table E2. Twelve are repeats of vbe_adder_3 and grover_5 pooled in Table E3. Section 3.7.1 reports them.

[F] The emulator jobs ran P0 on every circuit. P2 ran on every circuit except grover_5 and ladder_n6_r1 on the H2-Emulator, and grover_5 under P2 ran on H2-2E. P1 ran on the three barenco_tof circuits and the three ladder_n4 circuits of Table E1. [O] P1, P3 and P4 were otherwise left out because the count tables of Section 3 show them above P0 on ZZPhase count on every family A circuit and on 14 of 15, 14 of 15 and 15 of 15 family B1 circuits. Emulating them would spend HQCs to confirm a lower fidelity.

3. Results

3.1 Family A, Trotter spin chains

[F] Table A1. Mean ZZPhase count by n and pipeline, 36 circuits per n. "ZZ terms" is n_cx / 2 of the circuit as written, the number of two-body Pauli exponentials in the Trotter formula.

n	circuits	ZZ terms	P0ref	P0	P0L1	P1	P2	P3	P4
4	36	12.4	24.9	12.4	24.9	25.9	12.9	26.1	31.4
6	36	21.0	42.0	21.0	42.0	62.0	22.7	58.9	71.0
8	36	29.6	59.1	29.6	59.1	107.6	32.2	108.2	119.9
10	36	38.1	76.2	38.1	76.2	173.7	41.7	157.2	189.8
12	36	46.7	93.3	46.7	93.3	246.1	50.7	233.4	262.6

[F] Table A2. Each pipeline against P0 on the same circuit, 180 circuits. Ratios are averaged over circuits.

pipeline	better than P0	equal to P0	worse than P0	mean ZZ ratio	min ZZ ratio	max ZZ ratio	mean 2q-depth ratio	mean HQC ratio
P0ref	0	0	180	2.00	2.00	2.00	2.00	1.63
P0L1	0	0	180	2.00	2.00	2.00	2.00	1.63
P1	0	0	180	4.43	1.28	10.32	10.92	3.03
P2	0	120	60	1.04	1.00	1.19	1.25	1.03
P3	0	0	180	3.79	1.31	7.91	10.29	2.76
P4	0	0	180	4.90	1.44	10.36	11.88	3.33

[F] Table A3. P0 against the ZZ term count, by model and Trotter steps.

model	r	circuits	mean ZZ terms	mean n_2q P0	circuits with n_2q = ZZ terms
ladder	1	20	10.0	10.0	20
ladder	2	20	20.0	20.0	20
ladder	4	20	40.0	40.0	20
tfim	1	20	7.0	7.0	20
tfim	2	20	14.0	14.0	20
tfim	4	20	28.0	28.0	20
xxz	1	20	21.0	21.0	20
xxz	2	20	42.0	42.0	20
xxz	4	20	84.0	84.0	20

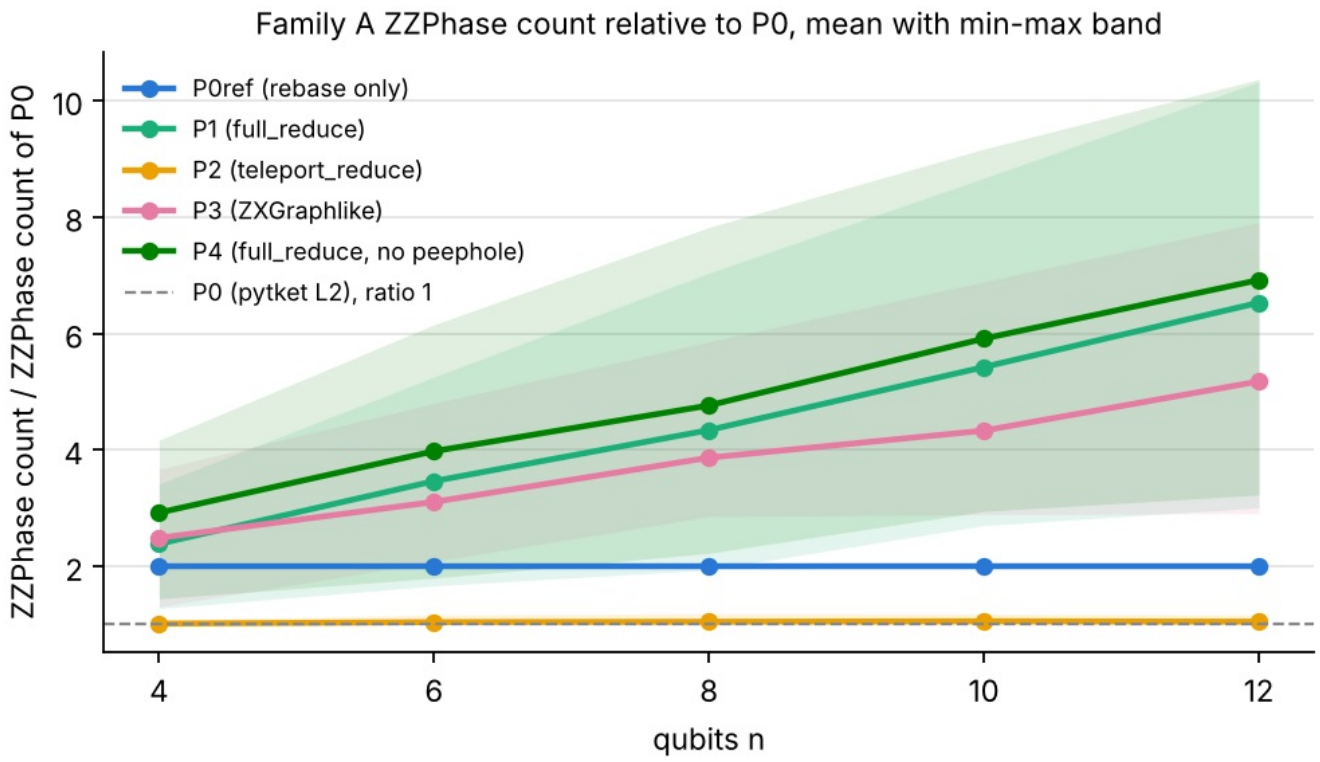


Figure 1. Family A. ZZPhase count of each pipeline divided by the P0 count on the same circuit, mean over the 36 circuits at each n with the min-max band shaded. The dashed line is P0 itself. Data: Table A2 and `results/family_A_counts.csv`.

`[F]` P0 compiles every one of the 180 circuits to exactly one ZZPhase per two-body term. No pipeline goes below that count on any circuit. P2 equals P0 on 120 circuits and is worse on 60, by at most a factor 1.19. P1, P3 and P4 are worse than P0 on all 180 circuits. The ratio to P0 runs from 1.28 to 10.32 for P1, from 1.31 to 7.91 for P3 and from 1.44 to 10.36 for P4. The mean ratio grows with n for each of the three, from 2.39 at n = 4 to 6.53 at n = 12 for P1, from 2.49 to 5.18 for P3 and from 2.92 to 6.92 for P4.

`[O]` One ZZPhase per two-body exponential is the natural count for this family and P0 reaches it. `[F]` No pipeline goes below the P0 count on any of the 180 circuits. `[O]` The ZX extraction pipelines add ZZPhase gates because `extract_circuit` resynthesises the graph with CNOTs that no longer follow the Trotter term structure.

3.2 Family B1, Clifford+T benchmarks

`[F]` Table B1. ZZPhase count per circuit and pipeline. "T after ZX" is the T-count of the PyZX circuit after the ZX stage of P1 and P2, which agree on every circuit. The hwb8 rows under P1 and P2 come from the 5400 s run described below.

circuit	n	T written	ZZ P0ref	ZZ P0	ZZ P0L1	ZZ P1	ZZ P2	ZZ P3	ZZ P4	P2 vs P0 %	T after ZX	note
tof_3	5	21	18	15	18	22	14	25	25	6.7	15	
barenco_tof_3	5	28	24	20	24	23	19	27	27	5.0	16	
mod5_4	5	28	28	24	28	25	22	20	29	8.3	8	
qft_4	5	69	46	36	46	43	36	64	57	0.0	67	
tof_4	7	35	30	25	30	31	23	40	38	8.0	23	
barenco_tof_4	7	56	48	40	48	41	38	54	54	5.0	28	
hwb6	7	105	116	99	116	130	94	189	144	5.1	75	
mod_mult_55	9	49	48	41	48	71	39	85	90	4.9	35	
tof_5	9	49	42	35	42	48	32	56	66	8.6	31	
barenco_tof_5	9	84	72	60	72	70	57	90	87	5.0	40	
grover_5	9	336	288	240	288	256	218	322	278	9.2	166	
vbe_adder_3	10	70	66	56	66	55	41	61	67	26.8	24	
mod_red_21	11	119	105	88	105	107	83	140	132	5.7	73	
gf2_pow_4_mult	12	112	99	83	99	260	83	277	290	0.0	68	
hwb8	12	5887	7118	5912	7118	8305	5587	11537	8461	5.5	3517	

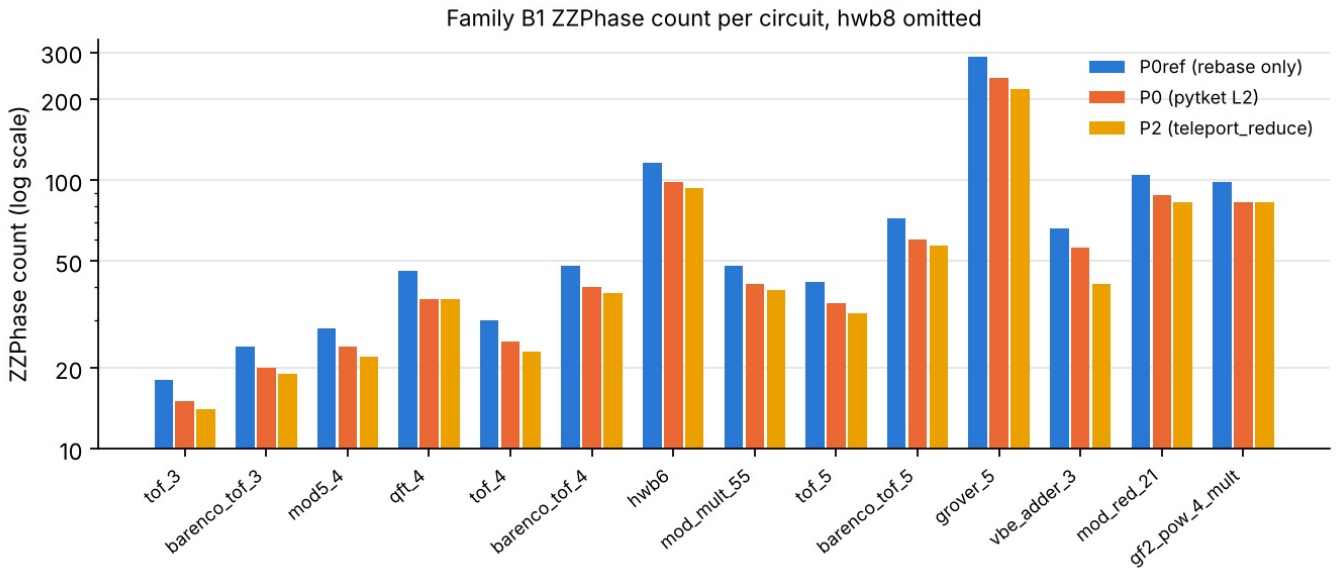


Figure 2. Family B1. ZZPhase count per circuit for P0ref, P0 and P2 on a log scale. Circuits are ordered by qubit number and then by T-count. hwb8 (5912 under P0) is omitted. Data: Table B1.

[F] On the 15 circuits P2 is below P0 on 13, equal on 2 (qft_4 and gf2_pow_4_mult) and above on none. The mean reduction against P0 is 6.9 %. The largest is vbe_adder_3, 56 to 41 ZZPhase gates, 26.8 %. P1 is above P0 on 14 of 15 and below on one (vbe_adder_3). P3 is above P0 on 14 of 15 and below on one (mod5_4). P4 is above P0 on 15 of 15.

[F] The hwb8 rows under P1 and P2 come from a separate run with a 5400 s limit (`results/family_B1_hwb8_long_counts.csv`, run log `results/family_B1_hwb8_long_run.log`), which completed P1 in 886 s and P2 in 844 s of wall time. P1 gives 8305 ZZPhase gates against 5912 under P0 and P2 gives 5587, a reduction of 5.5 %. Both leave the T-count at 3517, the P4 value. The two rows are merged into `results/family_B1_counts.csv`, and Table B1 and the counts in the previous paragraph include them. The pre-merge file is kept as `results/family_B1_counts_before_hwb8_merge.csv`.

[F] The T-counts after the ZX stage equal the PyZX column of Table 1 of arXiv:1903.10477 (Reference 3). This holds for all 13 family B1 circuits that table lists. It holds for the T-count as written and after phase teleportation. The check used arXiv:1903.10477v3 on 7 October (`results/literature_tcount_check.md`). grover_5 and qft_4 are not in that table. Their values 166 and 67 equal the T-counts of the optimised circuits shipped in the PyZX repository (circuits/optimized/grover_5_pyzx.qc and qft_4_pyzx.qc, commit e695d6d7).

- tof_3: 21 to 15
- mod5_4: 28 to 8
- vbe_adder_3: 70 to 24
- grover_5: 336 to 166

[0] Teleport reduction keeps the circuit skeleton and only moves phases, so the pytket pass that follows sees a circuit of the same shape with fewer non-Clifford phases and can merge more. Full reduction with extraction replaces the skeleton, and the extracted CNOT structure costs more two-qubit gates on H2 than the original.

[F] The certified T-count floors for the four n = 5 circuits are in Table C1 of Section 3.4. The floor is at or below the T-count after the ZX stage on all four, and the ratio floor / T after ZX is 25.00 % for barenco_tof_3 and mod5_4, 26.67 % for tof_3 and 55.22 % for qft_4.

[F] Neither the H2-Emulator rows of Section 3.7 nor the H2-2E rows of Section 3.7.1 resolve P2 against P0 on a single circuit at 1000 shots. At 4000 shots per pipeline the pooled H2-2E runs of Table E3 resolve P2 above P0 on vbe_adder_3 with z = +2.38. They do not resolve grover_5.

3.2.1 Family B1x, 13 to 20 qubits, count-only

[F] The 8 circuits of family B1x are unchecked. No unitary, statevector or distribution check was run on them and only the gate-set check applies. Every row of `results/family_B1x_counts.csv` carries the status `ok-unchecked`. [0] Their counts are reported here as a separate block and are not merged into the family B1 totals above. [F] Table B1x. ZZPhase count and two-qubit depth per circuit under P0ref, P0 and P2, with the T-count as written and the T-count after the ZX stage of P2. Circuits are ordered by n and then by T-count as written.

Count-only rows, status ok-unchecked on every row: no unitary, statevector or distribution check was run at this size, only the gate-set check. P2 vs P0 % = 100 (1 - ZZ P2 / ZZ P0).

file	n	ZZ P0ref	ZZ P0	ZZ P2	P2 vs P0 %	depth_2q P0	depth_2q P2	T written	T after ZX P2
rc_adder_6	14	88	76	73	3.9	48	43	77	47
csla_mux_3	15	76	63	63	0.0	31	32	70	62
gf2_pow_5_mult	15	154	129	129	0.0	68	84	175	115
ham15-low	17	236	204	198	2.9	145	124	161	97
ham15-med	17	534	427	379	11.2	344	266	574	212
gf2_pow_6_mult	18	221	185	185	0.0	86	90	252	150
tof_10	19	102	85	77	9.4	77	68	119	71

barenco_tof_10	19	192	160	152	5.0	145	130	224	100	

[F] On the 8 circuits P2 is below P0 on 5, equal on 3 (csla_mux_3, gf2_pow_5_mult and gf2_pow_6_mult) and above on none. The aggregate ZZPhase count over the 8 circuits is 1329 under P0 and 1256 under P2, a reduction of 5.5 %. The mean of the per-circuit reductions is 4.1 %. The largest reduction is ham15-med, 427 to 379 ZZPhase gates, 11.2 %. P0 is below P0ref on 8 of 8, with an aggregate of 1603 under P0ref against 1329 under P0. The two-qubit depth under P2 is below P0 on the 5 circuits where the count is below and above P0 on the 3 circuits where the count is equal. The ZX stage of P2 leaves no phase outside the pi/4 grid on any circuit and lowers the T-count from 1652 written to 854 over the 8 circuits.

[F] On rc_adder_6 the circuit as written has 93 CX gates and the P0ref compilation has 88 ZZPhase gates, so the rebase-only pipeline already removes 5 two-qubit gates before any optimisation. The same happens on csla_mux_3, 80 CX as written against 76 under P0ref. On the other 6 circuits the P0ref count equals the CX count as written. [F] pytket `RemoveRedundancies` applied alone to the written circuit leaves 91 CX on rc_adder_6 and 72 on csla_mux_3 (recomputed on 8 October), so the P0ref figures come from the whole stage sequence of P0ref and not from one cancellation rule. [0] The P0ref count is the right reference for the P0 and P2 reductions on these two circuits, and the CX count as written is not.

[0] The combined statement for the Clifford+T benchmarks is: checked family B1, 15 circuits, P2 below P0 on 13 and equal on 2. Count-only family B1x, 8 circuits, P2 below P0 on 5 and equal on 3. The two totals stay separate because the B1x rows have no exactness check.

3.3 Family B2, synthesised Haar unitaries

[F] Table B2. Means over 5 circuits per row. "T red. %" is the T-count reduction by the ZX stage against the circuit as written. "max target dev." is the largest 1 - |tr(U_target^dagger U_compiled)| / 2^n in the row.

n	eps	circuits	mean T written	T after ZX P1	T red. % P1	T after ZX P2	T red. % P2	T after ZX P3	T red. % P3	ZZ P0	ZZ P2	HQC P0	max target dev.
2	1e-05	5	1048.4	1024.4	2.3	1024.4	2.3	1037.6	1.0	3	3	16.6	3.2e-10
2	0.001	5	638.4	624.4	2.2	624.4	2.2	628.4	1.6	3	3	16.6	1.9e-06
3	1e-05	5	4220.8	4167.2	1.3	4167.2	1.3	4189.6	0.7	19	19	56.4	9.4e-10
3	0.001	5	2571.6	2490.8	3.1	2490.8	3.1	2522.8	1.9	19	19	56.4	7.1e-06

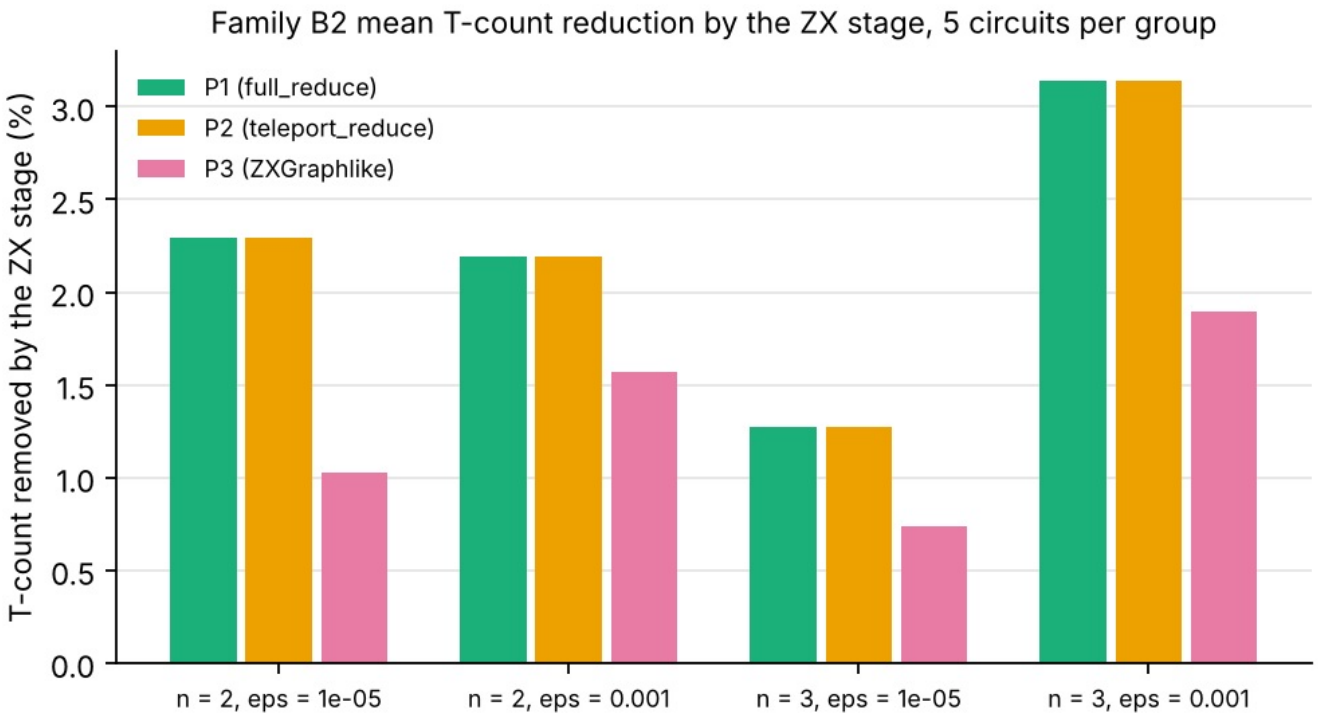


Figure 3. Family B2. Percentage of the written T-count removed by the ZX stage of P1, P2 and P3, mean over the 5 circuits in each (n, eps) group. P1 and P2 give the same T-count on every circuit. Data: Table B2.

[F] The mean raw T-counts match the barrier manuscript's Table E6 for the two rows it reports: n = 2 at eps = 1e-3 gives 638.4 here against 644 there, and n = 3 at eps = 1e-3 gives 2571.6 against 2570. The manuscript rows are means over 50 and 8 instances, this table over 5.

[F] The ZX stage removes between 1.3 % and 3.1 % of the T-count under P1 and P2, and between 0.7 % and 1.9 % under P3. The ZZPhase count after P0 is 3 at n = 2 and 19 at n = 3 for every circuit, and P2 leaves it unchanged.

[0] Synthesised circuits are the case where the fault-tolerant layer matters and the NISQ layer does not. The two-qubit count is fixed by the KAK or QSD skeleton. The whole cost is in the gridsynth words, which are near their floors by construction, and the certificate measures how near.

[F] The certified floor of every one of the 20 circuits is in Table C1 of Section 3.4. The floor divided by the T-count after the ZX stage averages 99.84, 99.79, 99.57 and 99.76 % over the four (n, eps) configurations in the row order of Table C1. The minimum over the 20

circuits is 99.28 %.

3.4 Certificates

[F] Table C1. The exactness-barrier certificate (barrier/certificate.py, Section 2.5) on the 20 family B2 circuits and the four family B1 circuits with n = 5, from results/certificates.csv. The 4^n x 4^n channel is small at these sizes. The floor is the least denominator exponent of the exact Pauli channel of the circuit as written. "T after ZX" is the T-count after PyZX teleport_reduce, which equals the full_reduce value on every row.

Per circuit. floor = lde of the exact Pauli channel (Corollary cert, Theorem C(c)), T after ZX = T-count after PyZX teleport_reduce (P2, equal to the full_reduce value on every row), achieved % = 100 (1 - T after ZX / T written).

family	file	n	T written	floor	T after ZX	floor / T after ZX %	achieved %
B2	haar_n2_eps1e-03_s1	2	662	651	652	99.85	1.51
B2	haar_n2_eps1e-03_s2	2	628	611	612	99.84	2.55
B2	haar_n2_eps1e-03_s3	2	638	630	630	100.00	1.25
B2	haar_n2_eps1e-03_s4	2	622	613	614	99.84	1.29
B2	haar_n2_eps1e-03_s5	2	642	612	614	99.67	4.36
B2	haar_n2_eps1e-05_s1	2	1046	1021	1022	99.90	2.29
B2	haar_n2_eps1e-05_s2	2	1046	1025	1026	99.90	1.91
B2	haar_n2_eps1e-05_s3	2	1038	1002	1004	99.80	3.28
B2	haar_n2_eps1e-05_s4	2	1052	1030	1036	99.42	1.52
B2	haar_n2_eps1e-05_s5	2	1060	1033	1034	99.90	2.45
B2	haar_n3_eps1e-03_s1	3	2566	2483	2496	99.48	2.73
B2	haar_n3_eps1e-03_s2	3	2544	2442	2452	99.59	3.62
B2	haar_n3_eps1e-03_s3	3	2570	2486	2490	99.84	3.11
B2	haar_n3_eps1e-03_s4	3	2588	2511	2520	99.64	2.63
B2	haar_n3_eps1e-03_s5	3	2590	2478	2496	99.28	3.63
B2	haar_n3_eps1e-05_s1	3	4208	4132	4144	99.71	1.52
B2	haar_n3_eps1e-05_s2	3	4218	4158	4162	99.90	1.33
B2	haar_n3_eps1e-05_s3	3	4224	4179	4188	99.79	0.85
B2	haar_n3_eps1e-05_s4	3	4248	4184	4192	99.81	1.32
B2	haar_n3_eps1e-05_s5	3	4206	4133	4150	99.59	1.33
B1	barenco_tof_3	5	28	4	16	25.00	42.86
B1	mod5_4	5	28	2	8	25.00	71.43
B1	qft_4	5	69	37	67	55.22	2.90
B1	tof_3	5	21	4	15	26.67	28.57

Family B2 by configuration, 5 circuits each.

n	eps	circuits	mean T written	mean floor / T after ZX %	min floor / T after ZX %	mean achieved %	mean collapse rate %
2	1e-03	5	638.4	99.84	99.67	2.19	1.25
2	1e-05	5	1048.4	99.79	99.42	2.29	1.34
3	1e-03	5	2571.6	99.57	99.28	3.14	1.86
3	1e-05	5	4220.8	99.76	99.59	1.27	0.80

[F] The invariant floor <= T after ZX holds on 24 of 24 rows, for the full_reduce and the teleport_reduce T-counts alike, and the two T-counts are equal on 24 of 24 rows. The certificate ran in at most 0.4 s per circuit at n = 2, at most 36.4 s at n = 3 and at most 38.5 s on the n = 5 family B1 circuits (wall_s in the CSV).

[F] On family B2 the configuration means of the floor divided by the T-count after the ZX stage are 99.84, 99.79, 99.57 and 99.76 % in the row order of Table C1. The minimum over the 20 circuits is 99.28 % (haar_n3_eps1e-03_s5, floor 2478 against 2496 after ZX) and the maximum is 100.00 % (haar_n2_eps1e-03_s3). The barrier manuscript calls this quantity lde / T_tel (caption of Table tab:e6 of arXiv:2608.22801). Its Table tab:e6 reports 99.79 +/- 0.30 with minimum 98.6 at (2, 1e-3) over 50 instances and 99.55 +/- 0.20 with minimum 99.3 at (3, 1e-3) over 8 instances. Its subsection on multi-qubit rigidity in the section on numerical methods and results states that the quantity averages 99.4 to 99.9 % and never falls below 96.9 % on any instance. All four configuration means here lie inside 99.4 to 99.9 % and every instance is above 96.9 %. The (n, eps) = (2, 1e-5) and (3, 1e-5) configurations have no counterpart in that table. The same subsection gives the collapse rate as 1.0 to 2.3 %. The collapse rate here is 1.25, 1.34, 1.86 and 0.80 % in the same row order. The last configuration sits below that range.

[0] The number a user reads from Table C1 is the gap between the two percentages. On family B2 the ZX stage removes 1.3 to 3.1 % of the written T-count. The certificate shows that at most 0.16 to 0.43 % of the remaining count is removable by any semantics-exact rewrite on the configuration means and at most 0.72 % on any single instance. The gridsynth words are within a fraction of a percent of their floor and no further exact optimiser effort on them is worth paying for.

[F] On the four family B1 circuits the floor is loose. For mod5_4 the floor is 2 while the ZX stage leaves 8 T gates, so the floor is 25.00 % of the achieved count, and the denominator exponent collapses at 19 of the 28 T gates in the certificate's step record (collapse_count in the CSV). The ratios on barenco_tof_3 and tof_3 are 25.00 % and 26.67 % and on qft_4 it is 55.22 %.

[0] The B1 floors are loose because these circuits are hand-written Toffoli networks, not synthesised ring elements. [F] The theorem bounds the T-count of every exact rewrite from below by the denominator exponent of the channel. On a gridsynth word of family B2 the exponent grows by one at all but 0.5 to 2.5 % of the T gates (collapse_rate in results/certificates.csv). [0] The bound is therefore tight there. In a Toffoli network most T gates cancel in the channel and the exponent stays small, so the bound stays well below what any known rewrite reaches. The gap between 2 and 8 on mod5_4 is either headroom for a better rewrite or slack in the bound, and the certificate does not say which. The manuscript's percentages are stated for synthesised circuits only and the B1 rows are not covered by them.

[F] Table C2 places the smallest T-count reported in the six papers for each B1 and B1x circuit beside our T-count after the ZX stage and the certified floor of Table C1. The values were copied from the tables of six papers fetched from arXiv on 8 October 2026 and are listed per source in `results/literature_tcounts.csv`. A published value enters the ancilla-free column only if the paper's circuit has our qubit count and our written T-count. AQ2024 lists no original T-count and takes its circuits from Amy's benchmark set (arXiv:2402.14396, "We take the circuits from Amy [36]", text before Table 2). Nam2018 lists no qubit count. For these two sources the match rests on the circuit name. Values from methods that add Hadamard ancillas sit in a separate column because the floor bounds circuits on the n circuit qubits only. grover_5 has no entry because the published Grover5 starts from 140 T gates and ours from 336.

[F] Per circuit of families B1 and B1x. T after ZX is the P2 T-count. best ancilla-free is the smallest T-count reported in the six papers on the same n qubits, taken over the columns of `results/literature_tcounts.csv` with ancilla_free = yes, same_circuit = yes and reported_error = no. best with ancillas is the smallest published T-count of a method that adds Hadamard ancillas. These methods are TODD and AlphaTensor-Quantum when its compiled qubit count exceeds n. The floor is the certified lde floor of Table C1 and exists for the four n = 5 circuits only. T after ZX - best is the part of the ZX output that a published ancilla-free method is known to remove. best - floor is the part that neither a published method nor the certificate has closed. Source keys: AMM2014 arXiv:1303.2042 Table 1, AM2016 arXiv:1601.07363 Table II, Nam2018 arXiv:1710.07345 Table 4, TODD2018 and TODDpart2018 arXiv:1712.01557 Tables I and II, KvdW2020 arXiv:1903.10477 Table 1, AQ2024 arXiv:2402.14396 Table 2. grover_5 has no same-circuit published value.

family	circuit	n	T written	T after ZX	best ancilla-free	source	best with ancillas	source (ancillas)	floor	T after ZX - best	best - floor
B1	barenco_tof_3	5	28	16	16	AM2016_RMmaj, AM2016_RMrec, AM2016_Tpar, AMM2014_Tpar, KvdW2020_PyZX, KvdW2020_PyZXTODD, Nam2018_H, Nam2018_L	13	AQ2024	4	0	12
B1	mod5_4	5	28	8	7	AQ2024, KvdW2020_PyZXTODD	16	TODD2018	2	1	5
B1	qft_4	5	69	67	67	AM2016_RMmaj, AM2016_RMrec, AM2016_Tpar, TODDpart2018	53	AQ2024	37	0	30
B1	tof_3	5	21	15	15	AM2016_RMmaj, AM2016_RMrec, AM2016_Tpar, AMM2014_Tpar, KvdW2020_PyZX, KvdW2020_PyZXTODD, Nam2018_L, TODDpart2018	13	AQ2024, TODD2018	4	0	11
B1	barenco_tof_4	7	56	28	28	AM2016_RMmaj, AM2016_RMrec, AM2016_Tpar, AMM2014_Tpar, KvdW2020_PyZX, KvdW2020_PyZXTODD, Nam2018_H, Nam2018_L	23	AQ2024		0	
B1	hwb6	7	105	75	71	AM2016_Tpar	51	AQ2024, TODD2018		4	
B1	tof_4	7	35	23	23	AM2016_RMmaj, AM2016_RMrec, AM2016_Tpar, AMM2014_Tpar, KvdW2020_PyZX, KvdW2020_PyZXTODD, Nam2018_L, TODDpart2018	19	AQ2024, TODD2018		0	
B1	barenco_tof_5	9	84	40	40	AM2016_RMmaj, AM2016_RMrec, AM2016_Tpar, AMM2014_Tpar, KvdW2020_PyZX, KvdW2020_PyZXTODD, Nam2018_H, Nam2018_L	33	AQ2024		0	
B1	grover_5	9	336	166							
B1	mod_mult_55	9	49	35	20	KvdW2020_PyZXTODD	17	AQ2024, TODD2018		15	
B1	tof_5	9	49	31	31	AM2016_RMmaj, AM2016_RMrec, AM2016_Tpar, AMM2014_Tpar, KvdW2020_PyZX, KvdW2020_PyZXTODD, Nam2018_L, TODDpart2018	25	AQ2024, TODD2018		0	
B1	vbe_adder_3	10	70	24	24	AM2016_RMmaj, AM2016_RMrec, AM2016_Tpar, AMM2014_Tpar, KvdW2020_PyZX, KvdW2020_PyZXTODD, Nam2018_L	19	AQ2024		0	
B1	mod_red_21	11	119	73	73	AM2016_RMmaj, AM2016_RMrec, AM2016_Tpar, AMM2014_Tpar, KvdW2020_PyZX, KvdW2020_PyZXTODD, Nam2018_H, Nam2018_L	51	AQ2024		0	
B1	gf2_pow_4_mult	12	112	68	39	AQ2024	54	TODD2018		29	
B1	hwb8	12	5887	3517	3501	KvdW2020_PyZXTODD				16	
B1x	rc_adder_6	14	77	47	47	AM2016_RMmaj, AM2016_RMrec, KvdW2020_PyZX, KvdW2020_PyZXTODD, Nam2018_H, Nam2018_L	37	AQ2024, TODD2018		0	
B1x	csla_mux_3	15	70	62	45	KvdW2020_PyZXTODD	39	AQ2024		17	
B1x	gf2_pow_5_mult	15	175	115	59	AQ2024	87	TODD2018		56	
B1x	ham15-low	17	161	97	97	AM2016_RMmaj, AM2016_RMrec, AM2016_Tpar, KvdW2020_PyZX, KvdW2020_PyZXTODD	73	AQ2024		0	
B1x	ham15-med	17	574	212	212	KvdW2020_PyZX, KvdW2020_PyZXTODD	162	TODD2018		0	
B1x	gf2_pow_6_mult	18	252	150	77	AQ2024	126	TODD2018		73	
B1x	barenco_tof_10	19	224	100	100	AM2016_RMmaj, AM2016_RMrec, AM2016_Tpar, AMM2014_Tpar, KvdW2020_PyZX, KvdW2020_PyZXTODD, Nam2018_H, Nam2018_L	83	AQ2024		0	
B1x	tof_10	19	119	71	71	AM2016_RMmaj, AM2016_RMrec, AM2016_Tpar, AMM2014_Tpar, KvdW2020_PyZX, KvdW2020_PyZXTODD, Nam2018_L, TODDpart2018	55	AQ2024, TODD2018		0	

[F] On the 4 circuits with a floor: T after ZX 106, best ancilla-free 105, floor 47. T after ZX - best = 1, best - floor = 58, sum 59. Best with ancillas on the same circuits: 95.

[F] On the four certified circuits the ZX stage leaves 106 T gates and the floor is 47. The best published ancilla-free T-counts sum to 105. Optimisers reported in the six papers therefore close 1 of the 59 gates between our count and the floor. That gate is on mod5_4, where PyZX with TODD and AlphaTensor-Quantum both reach 7 against our 8. On barenco_tof_3, tof_3 and qft_4 no ancilla-free method reported in the six papers goes below our count. The remaining 58 gates lie between the best published count and the floor and are closed by neither side.

[F] Our ZX stage matches the best published ancilla-free count on 14 of the 22 comparable circuits. On four of the other eight the better value comes from TODD applied after PyZX. On the three Galois-field multipliers it comes from AlphaTensor-Quantum. The hwb6 value of 71 is the T-par entry of arXiv:1601.07363 Table II, repeated as the input of arXiv:1712.01557 Table I. It conflicts with the text of arXiv:1601.07363 and with the 75 listed for T-par in arXiv:1903.10477. On the four certified circuits the best ancilla-free values of the six papers sum to 105 against 146 written T gates, a move of 41 gates. Beyond our ZX output the six papers lower the four certified circuits by one gate in total.

[0] The certificate cannot tell whether the 58 open gates are headroom or slack in the bound.

3.5 Noise proxy with H2-2E parameters

[F] Table A4. Local density-matrix proxy on family A at n <= 8 with 36 circuits per (n, pipeline) cell and 540 rows in `results/family_A_proxy_h22e.csv`. The simulator is `noise/dm_sim.py` with the published H2-2E emulator parameters from `noise/h2_params.py` (p1 = 7.3e-5, p2 = 1.29e-3, readout flips 9e-4 and 1.8e-3, p_init = 4e-5). The ideal distribution is the family reference `bench/family_A/<stem>.npy`.

Provisional local proxy. Density-matrix simulation with the published H2-2E emulator parameters (p1 = 7.3e-5, p2 = 1.29e-3, readout 9e-4 and 1.8e-3, p_init = 4e-5) and no leakage, crosstalk or dephasing, to be replaced by emulator runs. Mean Hellinger fidelity F_H of the noisy output distribution against the ideal one.

n	circuits per pipeline	F_H P0ref	F_H P0	F_H P1	F_H P2	F_H P3
4	36	0.9867	0.9922	0.9893	0.9919	0.9893
6	36	0.9785	0.9874	0.9735	0.9862	0.9743
8	36	0.9704	0.9826	0.9504	0.9807	0.9514

pipeline	mean F_H	mean TVD	mean n_2q	mean n_1q
P0ref	0.9785	0.0352	42.0	57.7
P0	0.9874	0.0210	21.0	40.1
P1	0.9710	0.0494	65.1	62.8
P2	0.9863	0.0221	22.6	42.0
P3	0.9717	0.0482	64.4	72.9

[F] The caption above is generated by `report/make_tables.py` for drafts v1 to v5 alike. [O] Its phrase "to be replaced by emulator runs" states the plan of the earlier drafts. [F] Table E1 of Section 3.7 holds 10 family A rows on four ladder circuits and Table E2 holds none. [O] The emulator runs therefore do not replace this proxy, which stays the primary noise data for family A.

[F] P0 and P2 have the highest fidelity in every one of the nine (model, n) cells of family A and P0 is above P0ref on 108 of 108 circuits. P0 and P2 differ by at most 0.0056 in any cell (xxz at n = 8), and per circuit P2 is above P0 on 43 and below on 65. P1 and P3 are below P0 and below P2 in every cell. The position of P0ref depends on the model. On ladder and tfim P0ref is above P1 and P3 in all six cells and on 36 of 36 circuits each. On xxz P0ref is below P1 in all three cells and below P3 at n = 4 and n = 6, because on xxz the extraction pipelines leave a ZZPhase count close to the P0ref count (30.3 against 42.0 at n = 4 under P1) while on ladder and tfim they leave 1.4 to 3.2 times the P0ref count. Over the 36 circuits at each n the ordering is P0 ~ P2 > P0ref > P1 ~ P3 at n = 6 and n = 8, and at n = 4 P1 and P3 (both 0.9893) sit above P0ref (0.9867).

[F] Table A5. The same proxy on the seven family B1 circuits with n <= 7, 35 rows in `results/family_B1_proxy_h22e.csv`. The ideal distribution is the noise-free output of the P0ref circuit. "support" is the number of outcomes with probability above 1e-12.

circuit	n	support	F_H P0ref	F_H P0	F_H P1	F_H P2	F_H P3	ZZ P0	ZZ P2
barenco_tof_3	5	1	0.9696	0.9739	0.9692	0.9747	0.9644	20	19
mod5_4	5	1	0.9647	0.9689	0.9661	0.9710	0.9705	24	22
qft_4	5	16	0.9641	0.9704	0.9623	0.9706	0.9527	36	36
tof_3	5	1	0.9759	0.9791	0.9691	0.9800	0.9655	15	14
barenco_tof_4	7	1	0.9430	0.9513	0.9451	0.9528	0.9316	40	38
hwb6	7	1	0.8751	0.8915	0.8521	0.8962	0.7977	99	94
tof_4	7	1	0.9614	0.9667	0.9567	0.9684	0.9464	25	23

[F] On family B1 P2 is above P0 on 7 of 7 circuits and P0 is above P0ref on 7 of 7. P1 is below P0 on 7 of 7 and P3 on 6 of 7. The exception is mod5_4, where P3 has the fewest ZZPhase gates of any pipeline (20 against 24 under P0). The largest fidelity gain of P2 over P0 is hwb6, 0.8915 to 0.8962 with 99 to 94 ZZPhase gates.

[F] A fit of -ln F_H = a n_2q with no intercept gives a = 1.18e-3 +/- 2.3e-5 on the 35 family B1 rows, with centred R^2 0.968, against the two-qubit error rate p2 = 1.29e-3 of the noise model. On the 540 family A rows the same fit gives a = 5.14e-4 +/- 1.3e-5 with centred R^2 0.612. Per model the slope is 2.28e-4 on ladder, 2.26e-4 on tfim and 7.58e-4 on xxz. The standard errors are from the residual variance and the R^2 is centred, 1 - SS_res / SS_tot about the mean of -ln F_H (`results/proxy_h22e_fits.log`, Section 4).

[F] The family B1 circuits have a single-outcome ideal distribution on 6 of 7 circuits (support 16 on qft_4), the xxz circuits have mean support 29.8 and the ladder and tfim circuits have mean support 107.8 out of 2^n. Adding the term u support to the family A fit -ln F_H = a n_2q + b n_1q gives u = -7.9e-5 per outcome. It raises the centred R^2 of that fit from 0.714 to 0.784.

[O] The slope follows the support of the ideal distribution. A depolarising error after a two-qubit gate moves probability off the ideal outcomes, and the Hellinger fidelity pays for it in full when the ideal distribution is a single outcome and only in part when the ideal distribution already covers most strings. The B1 slope close to p2 is the first case and the ladder and tfim slopes near 0.18 p2 are the second. A cost model that predicts F_H from gate counts therefore needs the support as a feature, which is the reason it enters Section 4.

[O] The proxy contains no leakage, crosstalk, idle dephasing or the RZZ angle scaling of the published H2-2E model, so it ranks pipelines by their ZZPhase count under the H2-2E gate rates and does not predict the emulator's absolute fidelity. The four family A circuits of Table E1 are too few to replace it.

3.6 Selene runs

[F] Tables S1 and S2. Runs of the compiled circuits on Quantinuum's local emulator Selene (`selene-sim` 0.3.4, route in Section 2.6), from `results/selene_noiseless.csv` (468 rows), `results/selene_depolarizing.csv` (68 rows) and `results/selene_depolarizing_full.csv` (468 rows). The noiseless runs cover the 144 family A circuits with n <= 10 and the 12 family B1 circuits with n <= 10 under P0, P1 and P2 at 2000 shots. The full depolarising run covers the same 156 circuits under P0, P1 and P2 at 2000 shots with `DepolarizingErrorModel` at the H2-2E rates p_1q = 7.3e-5, p_2q = 1.29e-3, p_meas = 1.35e-3 and p_init = 4e-5.

Noiseless runs (IdealErrorModel, Quest backend). TVD is against the exact output distribution of the compiled QASM, TVD_scale =

sqrt(support / (2 pi shots)).

family	circuits	pipelines	runs	shots per run	max TVD	max TVD / TVD_scale	flagged (TVD > 4 TVD_scale)	shots outside support
A	144	P0, P1, P2	432	2000	0.117	1.111	0	0
B1	12	P0, P1, P2	36	2000	0.080	1.121	0	0

Depolarising runs (DepolarizingErrorModel, p_1q = 7.3e-5, p_2q = 1.29e-3) against the density-matrix proxy, pipelines P0 and P2. z = (F_H Selene - mean F_H of 400 2000-shot samples of the proxy distribution) / their sd. gates_meas_init adds p_meas = 1.35e-3 and p_init = 4e-5, gates_only sets both to 0.

family	variant	runs	abs z <= 2	mean z	sd z	max abs z	mean F_H Selene - proxy exact	mean F_H Selene - proxy sampled
A	gates_meas_init	20	20	-0.13	0.82	1.73	-0.0075	-0.0007
A	gates_only	20	17	0.83	1.17	3.52	-0.0035	0.0033
B1	gates_meas_init	14	12	-0.50	1.18	2.81	-0.0029	-0.0027
B1	gates_only	14	10	1.24	1.11	2.43	0.0045	0.0046

[F] In the 468 noiseless runs the largest TVD between the sampled and the exact distribution is 0.117 and the largest ratio of TVD to its sampling scale sqrt(support / (2 pi N_shots)) is 1.121, so no run reaches the flag threshold of 4. No shot landed outside the exact support in any of the 468 runs. The support is smaller than 2^n on 276 runs, so a shot could have landed outside it there. The exact distribution of each compiled QASM file agrees with the family reference to 5.7e-11. The TVD divided by its asymptotic expectation for the actual distribution has median 0.93 and maximum 2.26 over the 468 runs. A pooled chi-square test against the exact multinomial on the 234 runs with support between 2 and 64 gives $p < 0.01$ on 4 runs and $p < 0.05$ on 13, against 2.3 and 11.7 expected under the null. The 4 runs were resampled with fresh seeds at 20000 shots (`results/selene_recheck.txt`) and none of the deviations persists.

[0] A wrong measure map or a wrong gate in the compiled QASM would put shots on zero-probability strings or push the TVD above its sampling scale. Neither happened on any of the 156 circuits under any of the three pipelines, so the compiled files run on Quantinuum's own toolchain and reproduce the distributions the local checks computed.

Depolarising runs on the full set (DepolarizingErrorModel, p_1q = 7.3e-5, p_2q = 1.29e-3, p_meas = 1.35e-3, p_init = 4e-5, 2000 shots). F_H is the Hellinger fidelity of the Selene samples to the ideal distribution. P0 > P1 and P2 > P1 count circuits with a strictly higher F_H.

family	n	circuits	mean F_H P0	mean F_H P1	mean F_H P2	P0 > P1	P2 > P1
A	4	36	0.9914	0.9880	0.9902	33	29
A	6	36	0.9832	0.9706	0.9820	35	32
A	8	36	0.9705	0.9410	0.9691	36	34
A	10	36	0.9517	0.8918	0.9501	36	36
B1	5	4	0.9722	0.9638	0.9739	4	3
B1	7	3	0.9328	0.9210	0.9308	2	2
B1	9	4	0.9121	0.8816	0.9109	4	4
B1	10	1	0.9410	0.9260	0.9350	1	1
all		156	0.9716	0.9459	0.9702	151	141

[F] Table S2. The full depolarising run has 468 runs on 156 circuits. The mean F_H over the 156 circuits is 0.9716 for P0, 0.9459 for P1 and 0.9702 for P2. P0 is strictly above P1 on 151 circuits and P2 is strictly above P1 on 141. Both P0 and P2 are strictly above P1 on 140 of the 156 circuits. The 16 exceptions are 14 family A circuits with n <= 8 and the B1 circuits barenco_tof_3 and barenco_tof_4.

[F] The mean of F_H(P2) minus F_H(P0) over the 156 circuits is -0.0013 and P2 is strictly above P0 on 67 circuits. The single-run sampling sd of F_H at 2000 shots has mean 0.0028 over the 345 runs with a proxy row. The mean difference is therefore 0.47 single-run sampling sd. The standard error of the mean difference over the 156 circuits is 0.0004.

[0] A single circuit's ordering of P2 against P0 is not resolved at 2000 shots. The ensemble mean favours P0 by 0.0013 in F_H, which is small next to the P1 deficit of 0.0256 against P0.

[F] 345 of the 468 runs have a density-matrix proxy row. The 123 runs without one are the family A circuits with n = 10 and the family B1 circuits with n = 9 and n = 10. On the 345 runs $|z| <= 2$ holds for 335, which is 97.1 %. The mean z is -0.001 and the population sd of z is 0.957 with largest $|z|$ 2.82. Independent draws from the proxy's sampling distribution give mean 0, sd 1 and 95.4 % within two sd. The mean of F_H under Selene minus the exact proxy F_H is -0.0051 and falls to +0.0002 against the proxy's F_H evaluated on 2000-shot samples.

[F] An earlier 68-run comparison on 17 circuits under P0 and P2 agreed within two sd on 32 of 34 runs in `gates_meas_init` and on 27 of 34 runs in `gates_only`, which leaves readout and initialisation noiseless.

[F] Selene's `DepolarizingErrorModel` was probed at p = 0.3 with 20000 shots per probe (`selene/calibrate_error_model.py`, `results/selene_calibration.txt`). Rz gates receive no noise. A single-qubit gate flips with probability 0.2051 +/- 0.0029 against 2p / 3 = 0.200 for a uniformly random non-identity Pauli. A ZZPhase leaves the 00 state with probability 0.2442 +/- 0.0030 against 12p / 15 = 0.240. Readout flips symmetrically at 0.306 and 0.304 and initialisation flips at 0.298. The conventions match `noise/dm_sim.py`, except that the proxy's readout flip is asymmetric and Selene's is one rate.

[0] Selene's `DepolarizingErrorModel` and the proxy share one gate-level convention, so their agreement checks the proxy's implementation and its handling of the measure map. The agreement does not test fidelity to H2-2E. Neither model contains the emission and leakage branch, the crosstalk or the duration-based dephasing of the published H2-2E model. The Selene runs are the first execution of the compiled circuits on a Quantinuum emulator in this submission and the emulator runs of Section 3.7 are the second.

[F] Wall time on the 2-core container was 465 s for the 468 noiseless runs and 68 s for the 68-run comparison, both with two processes. The per-run wall times of the 468 full depolarising runs sum to 1437 s. Every TVD and F_H value of the noiseless runs and the 68-run comparison was bit-identical between two complete runs of the driver. `results/selene.md` records one execution of the full

depolarising run.

3.7 Emulator runs

[F] Access. The team reached the Quantinuum platforms on 8 October 2026 through `aqora` 0.35.1 as team `learned_zx_rewriting`.

[F] Quotas. The Aqora forum topic "Emulator Cost" by Irfan Khan of Quantinuum sets 2 hours of H2-Emulator time per team and 10,000 HQC per team on H2-1E and H2-2E. The syntax checkers are free. The same topic states that teams have no usage tracking. No copy is in the repository.

[F] Syntax check. The H2-2SC syntax check of tof_3 under P0 completed as job `AFByb3ZpZGVySm9iAaEcdqFKfvGiVyt1pWCaEQ`. All 1000 shots returned the all-zero string and the job returned no cost. The raw payload is `results/emulator/nexus_H2-2SC/P0/tof_3.json`.

[F] Table E1. Nexus H2-Emulator runs with noise at 1000 shots, from `results/emulator_nexus_H2-Emulator.csv` (22 rows) and the counts in `results/emulator/nexus_H2-Emulator/`. Three jobs completed. The first ran tof_3 under P0 and P2 with a reported duration of 167 s. The second ran barenco_tof_3, barenco_tof_4 and barenco_tof_5 under P0, P1 and P2 and grover_5 under P0 in 953 s. The third ran ladder_n4_r1, ladder_n4_r2 and ladder_n4_r4 under P0, P1 and P2 and ladder_n6_r1 under P0 in 309 s. The job ledger is `results/emulator_jobs.csv`. `report/make_tables.py` recomputes every F_H from the stored counts and asserts agreement with the CSV to 1e-9.

Nexus H2-Emulator, noisy_simulation = True, 1000 shots per row. F_H is the Hellinger fidelity of the returned counts to the ideal distribution. sd F_H emulator is the sd of F_H over 4000 multinomial bootstrap resamples of the counts (seed 20261008). F_H proxy is the exact density-matrix proxy value of Section 3.5 and is empty where the proxy has no row. F_H Selene depol is the 2000-shot Selene DepolarizingErrorModel value of Section 3.6.

file	n	pipeline	n_2q	F_H emulator	sd F_H emulator	F_H proxy	F_H Selene depol
tof_3	5	P0	15	0.9860	0.0037	0.9791	0.9785
tof_3	5	P2	14	0.9810	0.0044	0.9800	0.9810
barenco_tof_3	5	P0	20	0.9770	0.0048	0.9739	0.9740
barenco_tof_3	5	P1	23	0.9760	0.0049	0.9692	0.9700
barenco_tof_3	5	P2	19	0.9660	0.0057	0.9747	0.9655
barenco_tof_4	7	P0	40	0.9500	0.0068	0.9513	0.9470
barenco_tof_4	7	P1	41	0.9420	0.0073	0.9451	0.9490
barenco_tof_4	7	P2	38	0.9660	0.0057	0.9528	0.9450
barenco_tof_5	9	P0	60	0.9280	0.0080		0.9305
barenco_tof_5	9	P1	70	0.9160	0.0088		0.9120
barenco_tof_5	9	P2	57	0.9130	0.0088		0.9255
grover_5	9	P0	240	0.7898	0.0131		0.8245
ladder_n4_r1_uniform_s1	4	P0	4	0.9993	0.0009	0.9992	0.9984
ladder_n4_r1_uniform_s1	4	P1	11	0.9927	0.0027	0.9954	0.9948
ladder_n4_r1_uniform_s1	4	P2	4	0.9981	0.0014	0.9992	0.9978
ladder_n4_r2_uniform_s1	4	P0	8	0.9993	0.0017	0.9995	0.9975
ladder_n4_r2_uniform_s1	4	P1	25	0.9901	0.0033	0.9933	0.9899
ladder_n4_r2_uniform_s1	4	P2	8	0.9947	0.0018	0.9995	0.9970
ladder_n4_r4_uniform_s1	4	P0	16	0.9957	0.0026	0.9998	0.9987
ladder_n4_r4_uniform_s1	4	P1	42	0.9912	0.0032	0.9970	0.9950
ladder_n4_r4_uniform_s1	4	P2	16	0.9924	0.0029	0.9998	0.9975
ladder_n6_r1_uniform_s1	6	P0	7	0.9963	0.0018	0.9986	0.9962

[F] Quota event. A fourth job of 10 family B1 circuits was rejected with `QuotaExceedException: Quota 'simulation' would be exceeded.` It held grover_5 under P1 and P2, hwb6 and mod5_4 under P0, P1 and P2 and mod_mult_55 under P0 and P1. The three completed jobs had reported 1429 s, which is 19.8 % of the 7200 s quota. The ledger records a duration of 823 s for the rejected job. [0] The cause of the gap between 1429 s and 7200 s is not known. One hypothesis is that the platform charges each circuit of a batch separately. The other is that it rejects a job on a duration pre-estimate before running it.

[F] Sampling error. The bootstrap sd of F_H at 1000 shots is 0.0037 to 0.0088 on the 11 B1 rows with a single ideal outcome and agrees with the binomial value sqrt(F_H (1 - F_H) / 1000) to 2.2 %. It is 0.0009 to 0.0033 on the 10 family A rows and 0.0131 on grover_5. The no-intercept fit of -ln F_H on n_2q over the B1 proxy rows gives 1.18e-3 per ZZ gate. P2 removes 1 to 3 ZZ gates against P0 on tof_3, barenco_tof_3, barenco_tof_4 and barenco_tof_5. The F_H gain expected from that slope is 0.0012 to 0.0033 on those four circuits. The sd of the measured P2 minus P0 difference is 0.0057 to 0.0119 on the same circuits. The expected gain is therefore 0.15 to 0.28 of one sd. The measured differences are -0.005, -0.011, +0.016 and -0.015 with z = -0.87, -1.48, +1.81 and -1.26 in the order above. All four lie within two sd of zero. At F_H = 0.955, the mean over the 11 single-outcome B1 rows, a one-gate gain reaches two sd of the difference only at 2.7e5 shots per pipeline.

[0] At 1000 shots the sampling error of F_H exceeds the fidelity difference expected from 1 to 3 fewer ZZ gates. The emulator data establish that the compiled circuits run correctly on the Quantinuum platform and that the team has used it. They do not resolve P2 against P0 on any single circuit. [F] In the Selene data of Table S2 F_H(P2) minus F_H(P0) has mean -0.0013 with standard error 0.0004 over the 156 circuits. Over the 12 B1 circuits the mean is -0.0009 with standard error 0.0016 and P2 is above P0 on 4. [F] On the B1 circuits Tables S2, E1 and E2 therefore do not resolve P2 against P0 in fidelity. Table E3 of Section 3.7.1 pools 4000 shots per pipeline. It resolves P2 above P0 on vbe_adder_3 with z = +2.38 and does not resolve grover_5. [0] The P2 against P0 comparison on H2 rests on the ZZ counts of Section 3.2 and the HQC counts of Section 1.2.

[F] Comparison with the proxy. On the 18 rows with a proxy value the mean of F_H emulator minus F_H proxy is -0.0007 and the mean absolute difference is 0.0042. The mean is -0.0031 on the 10 family A rows and +0.0022 on the 8 B1 rows. 15 of the 18 rows lie within two emulator sd of the proxy. The three rows outside are barenco_tof_4, ladder_n4_r2 and ladder_n4_r4, all under P2.

[F] Comparison with Selene. On all 22 rows the mean of F_H emulator minus F_H Selene is -0.0011 and the mean absolute difference is 0.0055. The sd of the difference combines the emulator sd with a Selene sd taken as the emulator sd times sqrt(1000 / 2000). 20 of

the 22 rows lie within two sd. The two rows outside are barenco_tof_4 under P2 and grover_5 under P0.

[F] On ladder_n4_r1, ladder_n4_r2 and ladder_n4_r4 the P2 circuit has the same ZZ count and two-qubit depth as P0 and no more PhasedX gates. The emulator F_H of P2 is below P0 on all three with z = -0.68, -1.88 and -0.83. [O] The proxy predicts equal F_H for these pairs. The published H2-2E noise model has angle-dependent two-qubit errors and dephasing that the proxy lacks. This report does not establish whether the H2-Emulator uses that model, and the difference is not resolved at 1000 shots.

3.7.1 H2-2E

[F] Runs. vbe_adder_3, hwb6 and grover_5 ran on the Nexus H2-2E emulator under P0 and P2 with `noisy_simulation = True` at 1000 shots. Each of the six circuits went in its own job. All six jobs completed on 8 October 2026 and are the `nexus:H2-2E` rows of `results/emulator_jobs.csv`. The job ids in the order P0 then P2 are the following.

- [F] vbe_adder_3: `AFByb3ZpZGVySm9iAaEcwoXVdfGm_ordi_7h5w`, `AFByb3ZpZGVySm9iAaEcyAXPe_KMnSd2cLrgKg`
- [F] hwb6: `AFByb3ZpZGVySm9iAaEczGiEcIO6R60fhgRfMw`, `AFByb3ZpZGVySm9iAaEc2Kp3dxCmIno2DMeWtA`
- [F] grover_5: `AFByb3ZpZGVySm9iAaEc31OVdBG5qZ8C3KUGnQ`, `AFByb3ZpZGVySm9iAaEc47DrcSKGODVcQ3R9uw`

[F] Cost. The HQC estimates stored with the six jobs sum to 1840.0 HQC. With the twelve repeat jobs described below the 18 H2-2E jobs sum to 5933.2 HQC. That is 59.3 % of the 10,000 HQC quota per team on H2-1E and H2-2E. The cost fields returned by the platform hold a duration and no HQC value. The charged amount is therefore not recorded in the repository.

[F] Table E2. The six rows of `results/emulator_nexus_H2-2E.csv` with the counts in `results/emulator/nexus_H2-2E/`. `submit/h22e_stats.py` recomputes every F_H from the counts. `report/make_tables.py` asserts agreement with the CSV and with the JSON payloads to 1e-9.

[F] Nexus H2-2E, noisy_simulation = True, 1000 shots per row, one job per row. F_H is the Hellinger fidelity of the returned counts to the ideal distribution. sd F_H is the binomial value sqrt(F_H (1 - F_H) / 1000) on vbe_adder_3 and hwb6, whose ideal output is a single bit string. On grover_5 it is the sd over 4000 multinomial bootstrap resamples of the counts (seed 20261008). HQC estimate is the formula value stored with each job. F_H proxy is the exact density-matrix proxy value of Section 3.5 and is empty where the proxy has no row (n > 7). F_H Selene depol is the 2000-shot Selene DepolarizingErrorModel value of Section 3.6.

file	n	pipeline	n_2q	HQC estimate	F_H	sd F_H	F_H proxy	F_H Selene depol
vbe_adder_3	10	P0	56	151.6	0.9540	0.0066		0.9410
vbe_adder_3	10	P2	41	119.0	0.9590	0.0063		0.9350
hwb6	7	P0	99	243.0	0.9240	0.0084	0.8915	0.8845
hwb6	7	P2	94	232.6	0.9090	0.0091	0.8962	0.8805
grover_5	9	P0	240	565.6	0.8652	0.0114		0.8245
grover_5	9	P2	218	528.2	0.8728	0.0112		0.8315

[F] Paired differences. F_H(P2) minus F_H(P0) is +0.0050 with sd 0.0091 and z = +0.55 on vbe_adder_3. It is -0.0150 with sd 0.0124 and z = -1.21 on hwb6. It is +0.0076 with sd 0.0160 and z = +0.48 on grover_5. All three lie within two sd of zero.

[F] Predicted gain. The no-intercept fit of -ln F_H on n_2q over the 35 family B1 proxy rows gives a = 1.1801e-3 per ZZ gate. `submit/h22e_stats.py` refits it and checks it against the value 1.1801e-3 printed at line 77 of `results/proxy_h22e_fits.log`. P2 removes 15, 5 and 22 ZZ gates against P0 on vbe_adder_3, hwb6 and grover_5. The predicted F ratio exp(a dN_2q) is 1.0179, 1.0059 and 1.0263 on the three circuits. Applied to the measured P0 value it gives predicted gains of 0.0170, 0.0055 and 0.0228. These are 1.87, 0.44 and 1.42 sd of the measured difference. The measured differences lie below the predicted gains by 1.32, 1.66 and 0.95 sd.

[F] Shots for a 2-sd resolution. The sd of the difference scales as one over the square root of the shot count. The predicted gain equals two sd of the difference at 1147 shots per pipeline on vbe_adder_3, 20462 on hwb6 and 1974 on grover_5.

[F] Comparison with the proxy. Only hwb6 has a proxy row because the proxy stops at n = 7. The emulator F_H of hwb6 lies 3.88 sd above the proxy under P0 and 1.41 sd above it under P2. The proxy predicts a P2 gain of +0.0047 on hwb6.

[F] Comparison with Selene. The emulator F_H exceeds the 2000-shot Selene depolarising value on all six rows. The excess runs from 0.0130 to 0.0413 with z from 1.60 to 3.85. Five of the six rows lie more than two sd above Selene. The sd combines the emulator sd with a Selene sd taken as the emulator sd times sqrt(1000 / 2000) as in Section 3.7.

[F] Comparison with the H2-Emulator. grover_5 under P0 also ran on the H2-Emulator in Table E1 with F_H 0.7898. Its H2-2E value 0.8652 is 4.33 sd higher. The sd combines the Table E1 sd 0.0131 with the Table E2 sd 0.0114. `submit/h22e_stats.py` prints this z.

[F] Repeat runs. vbe_adder_3 and grover_5 were submitted three more times to H2-2E under P0 and P2. The repeats used the compiled circuits, options and shot count of Table E2. Each repeat went in its own job. The 12 completed repeat jobs are the `nexus:H2-2E` rows of `results/emulator_jobs.csv` whose result paths lie under `results/emulator/nexus_H2-2E_r2/`, `_r3/` and `_r4/`. Their stored HQC estimates sum to 4093.2 HQC. With the six jobs of Table E2 the 18 H2-2E jobs sum to 5933.2 HQC. That is 59.3 % of the team quota of 10000 HQC. Each of the four cells now holds 4000 shots from four runs of 1000. hwb6 has the original run only and is not pooled.

[F] Nexus H2-2E, noisy_simulation = True, 1000 shots per run, one job per run. runs counts the original run of Table E2 and the completed repeats. pooled F_H is the Hellinger fidelity of the summed counts to the ideal distribution. sd is binomial on vbe_adder_3 and the sd over 4000 multinomial bootstrap resamples of the pooled counts on grover_5 (seed 20261008). The per-run F_H range is the smallest and largest F_H of single runs.

file	pipeline	runs	shots	n_2q	pooled F_H	sd	per-run F_H range
vbe_adder_3	P0	4	4000	56	0.9520	0.0034	0.9470 to 0.9560
vbe_adder_3	P2	4	4000	41	0.9628	0.0030	0.9590 to 0.9680
grover_5	P0	4	4000	240	0.8704	0.0053	0.8482 to 0.8652
grover_5	P2	4	4000	218	0.8711	0.0053	0.8400 to 0.8728

[F] vbe_adder_3 F_H(P2) - F_H(P0) +0.0108, sd 0.0045, z +2.38, p 0.017, predicted 0.0170, measured - predicted -1.38 sd. grover_5 F_H(P2) - F_H(P0) +0.0007, sd 0.0075, z +0.10, p 0.92, predicted 0.0229, measured - predicted -2.94 sd. Homogeneity p: vbe_adder_3

P0 0.80, vbe_adder_3 P2 0.70, grover_5 P0 0.91, grover_5 P2 0.68. H2-2E HQC estimate of the 18 completed jobs 5933.2, 59.3 % of 10000.

[F] Pooled numbers. `submit/h22e_pooled.py` computes F_H of each run and of the summed counts. On vbe_adder_3 the per-run F_H lies between 0.9470 and 0.9560 under P0 and between 0.9590 and 0.9680 under P2. A chi-square test on the per-run success and failure counts gives p = 0.80 under P0 and p = 0.70 under P2. The pooled F_H is 0.9520 with sd 0.0034 under P0 and 0.9628 with sd 0.0030 under P2. On grover_5 the per-run F_H lies between 0.8482 and 0.8652 under P0 and between 0.8400 and 0.8728 under P2. The homogeneity test there is a chi-square test on the per-run counts of the 64 support outcomes plus one bin for all other outcomes. It gives p = 0.91 under P0 and p = 0.68 under P2. The pooled F_H is 0.8704 with sd 0.0053 under P0 and 0.8711 with sd 0.0053 under P2. No cell rejects homogeneity at the 0.05 level. The pooled F_H of grover_5 under P0 exceeds every single-run value. Finite sampling lowers the measured F_H of a spread distribution and the summed counts sample it more finely. F_H(P2) minus F_H(P0) is +0.0108 with sd 0.0045 on vbe_adder_3. Its z is +2.38 and its two-sided p is 0.017. On grover_5 the difference is +0.0007 with sd 0.0075. Its z is +0.10 and its two-sided p is 0.92. The Holm adjustment of the two p values over the two circuits gives 0.035 on vbe_adder_3 and 0.92 on grover_5. Applied to the pooled P0 value the slope a = 1.1801e-3 used above predicts a gain of 0.0170 on vbe_adder_3 from 15 fewer ZZ gates. It predicts 0.0229 on grover_5 from 22 fewer. The measured difference lies 1.38 sd below the prediction on vbe_adder_3 and 2.94 sd below it on grover_5.

[O] On vbe_adder_3 the pooled data resolve the difference at 4000 shots per pipeline. P2 is above P0 by 0.0108 with z = +2.38. The measured difference lies within 2 sd of the gain predicted by the slope. On grover_5 the difference is not resolved at 4000 shots per pipeline. Its z is +0.10. The measured difference lies more than 2 sd below the gain predicted by the slope. After Holm adjustment over the two circuits the difference on vbe_adder_3 stays below p = 0.05.

[F] The single H2-2E runs do not resolve P2 against P0 at 1000 shots on any of the three circuits. At 4000 shots per pipeline the pooled runs resolve P2 above P0 on vbe_adder_3 with z = +2.38 and a Holm-adjusted p of 0.035. They do not resolve grover_5, where z is +0.10. They agree with the proxy within sampling error on hwb6 under P2 and not under P0. The emulator F_H lies above the proxy and above Selene on every row where either exists. At the predicted gains a 2-sd resolution needs 1147 shots per pipeline on vbe_adder_3 and 1974 on grover_5. On hwb6 it needs 20462. At 4000 shots the grover_5 difference lies 2.94 sd below its predicted gain.

[O] The H2-2E runs confirm execution and HQC accounting on the Quantinuum H2-2E emulator. One reading is that H2-2E applies less error to these B1 circuits than the depolarising model at the published rates. This report does not test that reading. vbe_adder_3 is the one circuit on which the emulator measures a fidelity gain of P2 over P0. The gain stays below p = 0.05 after the Holm adjustment over two circuits. On grover_5 the 22 fewer ZZ gates show no fidelity gain at 4000 shots and the slope prediction is not met.

4. Learned cost model

[F] Version 1 of this section was a protocol with no trained model. Version 2 trains the model on the local proxy fidelities of Section 3 and reports its validation. Every number in Tables M1 and M2 is produced by `report/make_tables.py` from `results/cost_model_metrics.csv` and `results/cost_model_coefficients.csv`, which are written by `model/cost_model.py`.

[F] Data. The training set is the proxy output of `noise/run_proxy.py` with the published H2-2E emulator defaults p1 = 7.3e-5, p2 = 1.29e-3, readout flip rates 9e-4 and 1.8e-3 and p_init = 4e-5. Family A contributes 540 rows, the 108 Trotter circuits with n in {4, 6, 8} under the five pipelines P0ref, P0, P1, P2 and P3. Family B1 contributes 35 rows, the 7 benchmark circuits with n in {5, 7} under the same five pipelines. F_H is the Hellinger fidelity of the exact noisy output distribution against the ideal one, with no shot sampling. [O] These are proxy numbers from a density-matrix model that omits leakage, crosstalk and duration-based dephasing. The script reruns unchanged on emulator CSVs with the same columns. [F] The model was not refit on the 22 H2-Emulator rows or the six H2-2E rows of Sections 3.7 and 3.7.1.

4.1 Target and features

[F] The target is y = -ln F_H. For a product of independent small error channels F_H is close to exp(-sum_k p_k N_k), so y is linear in the gate counts to first order. The seven features are read from the compiled circuit.

- [F] n_2q is the ZZPhase and ZZMax count. It carries the two-qubit fault probability p2.
- [F] n_1q is the PhasedX count. It carries p1. Rz is virtual and noiseless and is not counted.
- [F] n_qubits is the number of measured qubits. Each carries one readout and one initialisation fault.
- [F] depth_2q is the two-qubit depth. It stands in for idle error because the proxy has no duration model.
- [F] support_size is the number of bitstrings with ideal probability above 1e-12.
- [F] log2_support is the base-2 logarithm of support_size.
- [F] n_2q_offsupport = n_2q (1 - support_size / 2^n) is the interaction term.

[O] The interaction is justified by the metric. A Pauli fault after a gate moves a fraction of the output mass to other bitstrings. When the ideal distribution covers most of the 2^n strings the displaced mass lands on strings that already carry ideal weight and the Hellinger overlap barely changes. When the ideal output is one string every fault removes its mass from the support and the full per-gate error appears in y. The factor (1 - support_size / 2^n) is the fraction of the output space outside the support and is the first-order estimate of the lost mass. [F] The two families sit at opposite ends of this factor. The mean support fraction support_size / 2^n is 0.758 in family A and 280 of its 540 rows have full support. In family B1 the mean support fraction is 0.088 and the three values that occur are 0.0078, 0.031 and 0.5.

4.2 Models

[F] Four models are fitted to y on the same feature matrix. The physics baseline y = p2 n_2q + p1 n_1q + (p_meas_avg + p_init) n_qubits has no fitted parameter, with p_meas_avg + p_init = 1.39e-3 the mean of the two readout flip rates plus the initialisation fault. The OLS model is least squares on the seven features without intercept, since an empty circuit has y = 0 and the n_qubits column carries the SPAM floor. The ridge model scales each column to unit RMS on the training set and applies penalty lambda = 1 to the scaled coefficients, again without intercept. The gradient-boosted model is sklearn `GradientBoostingRegressor` with 300 trees of

depth 3, learning rate 0.05, subsample 0.8 and `random_state=0`. [O] The three fitted models are ordered by capacity. OLS tests whether a linear correction of the physics baseline is enough. Ridge tests whether the OLS coefficients are stable under shrinkage. The boosted model is the learned component and tests whether a nonlinear fit buys anything beyond the linear one.

4.3 Validation splits

[F] Three kinds of split are reported. Leave-one-family-out trains on A and tests on B1 and the reverse. Leave-one-model-out trains on two of the three Trotter models of A and tests on the third. Five-fold cross-validation on the pooled 575 rows is run twice. In the plain split cv5 the rows are shuffled into folds with seed 0. In the grouped split cv5g the folds are drawn over circuit classes, where a class is the file name with the disorder-seed suffix removed, so that all pipelines and all seeds of a class are in the same fold.

[F] The grouped split exists because the seeds of one circuit class are near-duplicate rows. 27 of the 54 classes of A have three disorder seeds. Within one class and one pipeline the three seeds have identical n_2q, identical depth_2q and identical support_size in 130 of the 135 three-seed class-pipeline groups, and their F_H values lie within 0.0056 of each other. In these 130 groups a random fold puts a near-copy of a test row in the training set unless all three seeds fall in the same fold. Its score therefore measures interpolation between pipelines and seeds of a circuit already seen. [F] The effect is visible in Table M1. The boosted model scores R^2 0.990 on cv5 and 0.971 on cv5g on the pooled rows, and 0.923 against 0.830 on the B1 rows. [O] The grouped split is the one that answers the question a user asks, which is the fidelity of a circuit the model has not seen, and it is the split used for the calibration figure and the permutation importance.

4.4 Results

[F] Table M1. R^2 of y and mean absolute error of F_H on the test set of each split, one cell per model. R^2 is taken about the mean of the test y and is negative for a model worse than that mean. n is the number of test rows.

split	train	test	n	physics	ols	ridge	gbr
lofo	A	B1	35	0.785 / 0.0151	0.560 / 0.0252	0.840 / 0.0150	0.805 / 0.0083
lofo	B1	A	540	-3.513 / 0.0430	-261.894 / 0.1971	-0.605 / 0.0259	0.326 / 0.0184
lomo	A minus ladder	A ladder	180	-34.797 / 0.0489	0.399 / 0.0058	0.410 / 0.0060	0.751 / 0.0030
lomo	A minus tfim	A tfim	180	-33.455 / 0.0424	0.637 / 0.0044	0.522 / 0.0047	0.843 / 0.0023
lomo	A minus xxz	A xxz	180	-2.061 / 0.0376	-1.272 / 0.0350	0.541 / 0.0159	-1.035 / 0.0353
cv5	A+B1 (4/5)	A+B1 (1/5)	575	-2.739 / 0.0413	0.944 / 0.0050	0.944 / 0.0050	0.990 / 0.0009
cv5g	A+B1 (4/5)	A+B1 (1/5)	575	-2.739 / 0.0413	0.938 / 0.0051	0.938 / 0.0052	0.971 / 0.0017
cv5g	A+B1 (4/5)	A rows of fold	540	-3.513 / 0.0430	0.937 / 0.0049	0.935 / 0.0050	0.990 / 0.0014
cv5g	A+B1 (4/5)	B1 rows of fold	35	0.785 / 0.0151	0.918 / 0.0090	0.926 / 0.0079	0.830 / 0.0074

[F] Table M2. Coefficients of y per unit of each feature. The physics column is fixed by the emulator parameters. The OLS columns give the coefficient and its t statistic for the three training sets. The ridge column is the shrunk coefficient on the pooled set. The last column is the permutation importance of the boosted model, the mean increase of out-of-fold mean squared error of y when that feature is shuffled in the test fold of cv5g, with the standard deviation over 5 folds times 10 shuffles. [O] The OLS standard errors assume independent homoscedastic residuals. The five pipeline rows of one circuit are not independent, so the t statistics are optimistic and only their ranking is used below.

feature	physics	OLS A (t)	OLS B1 (t)	OLS A+B1 (t)	ridge A+B1	GBR importance (sd)
n_2q	1.29e-03	3.73e-04 (11.8)	-4.77e-05 (-0.2)	2.89e-04 (8.2)	2.56e-04	5.95e-05 (1.2e-05)
n_1q	7.30e-05	7.38e-06 (0.4)	8.08e-05 (0.9)	4.65e-05 (2.2)	5.86e-05	2.91e-05 (4.0e-05)
n_qubits	1.39e-03	8.69e-03 (17.0)	-5.57e-05 (-0.1)	3.81e-03 (15.0)	3.49e-03	2.89e-06 (1.6e-06)
depth_2q		-1.90e-04 (-6.2)	1.68e-04 (1.0)	-1.40e-04 (-4.2)	-1.10e-04	4.13e-05 (2.2e-05)
support_size		2.85e-06 (0.6)	4.57e-03 (1.4)	-5.03e-06 (-0.9)	-4.99e-06	2.70e-06 (2.3e-06)
log2_support		-9.12e-03 (-16.2)	-1.71e-02 (-1.4)	-3.91e-03 (-13.6)	-3.57e-03	1.91e-06 (2.1e-06)
n_2q_offsupport		4.97e-04 (24.7)	9.81e-04 (4.1)	6.19e-04 (30.3)	6.27e-04	1.47e-03 (7.0e-04)

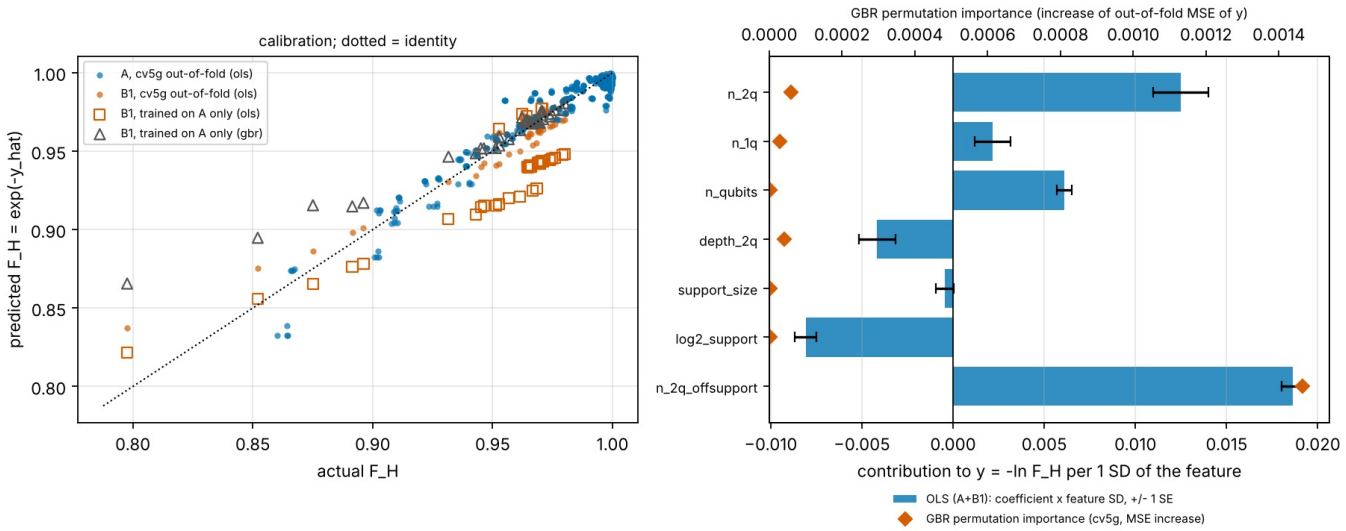


Figure 4. Learned cost model on the proxy data. Left, predicted F_H = exp(-y_hat) against actual F_H. Filled points are the out-of-fold cv5g predictions of the OLS model for family A in blue and family B1 in orange. Hollow squares are the B1 predictions of the OLS

model trained on A only and hollow triangles those of the boosted model trained on A only. The dotted line is the identity. Right, the OLS coefficients on the pooled set multiplied by the standard deviation of each feature with 1 SE bars on the lower axis, and the permutation importance of the boosted model as diamonds on the upper axis. Data: Tables M1 and M2, `results/fig_cost_model.png`.

[F] In the left panel 24 of the 29 hollow squares for the B1 rows with actual F_H between 0.94 and 0.98 lie below the identity line. The A-trained OLS model under-predicts the fidelity of those 24 rows. The hollow triangles of the A-trained boosted model lie closer to the line on the same circuits. The filled B1 points of the grouped cross-validation lie on or near the line over the full range from 0.80 to 0.98.

4.5 What the data support

[F] Across families the transfer is one-directional. Trained on A, the boosted model predicts B1 with R^2 0.805 and MAE(F_H) 0.0083. The unfitted physics baseline gives R^2 0.785 and MAE(F_H) 0.0151 on the same 35 rows. Ridge reaches R^2 0.840 with MAE(F_H) 0.0150 and OLS reaches 0.560 with 0.0252. Fitting on A lowers the B1 error of the best model by 0.0067 in MAE(F_H), a factor 1.81 below the baseline. Trained on B1, no model predicts A. The best R^2 in that direction is 0.326 for the boosted model and the other three are negative. [O] Seven circuits under five pipelines are 35 rows spanning three support fractions and cannot pin down seven coefficients, which the OLS B1 column of Table M2 shows directly. The only coefficient with |t| above 2 on B1 is n_2q_offsupport at 4.1. [O] The one direction that works rests on 7 test circuits and does not establish generalisation.

[F] Within family A the models interpolate across circuit classes. On the grouped split the A rows score R^2 0.937 for OLS and 0.990 for the boosted model. Holding out a whole Trotter model is harder. The boosted model reaches R^2 0.751 on ladder and 0.843 on tfim and fails on xxz at -1.035, where ridge is the only model above zero at 0.541. [F] The xxz circuits have the largest mean n_2q of the three models, 60.7 against 37.3 for ladder and 31.0 for tfim, and 4 of the 180 xxz rows exceed the largest n_2q of the other two. [O] The cause of the failure on xxz is not identified.

[F] The physics baseline fails on A in every split. Its R^2 on the 540 A rows is -3.513 with MAE(F_H) 0.0430, against 0.785 and 0.0151 on B1. [O] The baseline charges the full p2 per two-qubit gate. On A, where 280 of 540 rows have full support, a fault displaces mass onto strings that already carry ideal weight and the Hellinger fidelity loses far less than p2 per gate. On B1 the ideal output is one string on 6 of 7 circuits. There every fault that changes the output string leaves the support and the baseline is close to right.

[F] The dominant term in every fitted model is n_2q_offsupport. On the pooled set its OLS t statistic is 30.3, the largest in Table M2, and the next are n_qubits at 15.0 and log2_support at -13.6. Its permutation importance for the boosted model is 1.47e-3, and the second largest is n_2q at 5.95e-5, a factor 24.6 lower. In the right panel of Figure 4 its bar is the longest and its diamond is the only one away from zero. [F] The pooled OLS coefficients give a per-gate cost of n_2q that moves from 2.89e-4 at full support to 2.89e-4 + 6.19e-4 = 9.08e-4 at a one-string output, against the emulator value p2 = 1.29e-3.

[O] Verdict. What the data support across families is the physics baseline plus a support correction, not a learned model. The one quantity that transfers is the per-two-qubit-gate cost scaled by the fraction of the output space outside the ideal support. [F] The boosted model trained on A cuts the B1 error of the baseline by a factor 1.8 and OLS on the same features does worse than the baseline. [O] On these 35 rows the gain comes from the nonlinear fit and not from the feature set alone. [F] The gain does not survive the reverse direction. The grouped cross-validation ranks ridge above the boosted model on the B1 rows at 0.926 against 0.830 and the leave-one-model-out split on xxz ranks ridge first as well. The form these data support is y = a n_2q (1 - support_size / 2^n) + b n_2q + SPAM with two fitted parameters. The refit on emulator data will score that form beside the boosted model on held-out circuits, and the boosted model stays in the submission as the learned component only if it wins there.

4.6 Status and use

[F] All numbers in this section come from the local proxy with the published H2-2E parameters. The proxy has no shot noise, no leakage, no crosstalk and no duration-based dephasing. The emulator F_H of Sections 3.7 and 3.7.1 was measured at 1000 shots and does not enter this section. [O] Shot noise adds a floor to y that none of the seven features carries, so the refit on emulator data will need either an intercept or a shot-count term, and the coefficients of Table M2 will change.

[O] The model has two uses in this submission. Before HQCs are spent it ranks the five pipelines of a circuit by predicted F_H from their compiled counts, so that only the pipeline the model prefers is run. The rank correlation between predicted and measured F_H across the pipelines of a circuit is the metric that matters for that use. [F] That rank correlation was not computed on the emulator rows of Sections 3.7 and 3.7.1. [O] After the run it predicts the fidelity of any further circuit from its compiled counts and support size, which is what a user who has not yet run the circuit can supply. The support size is computed from the ideal distribution and is available for every circuit in this benchmark because the ideal outputs are stored, and it is not available for a circuit too large to simulate. A cost signal for a rewriting search is future work and is not claimed here.

4.7 Refit on Selene depolarising data

[F] The cost model of Sections 4.1 to 4.3 was refitted without code changes on the Selene depolarising runs of Section 3.6. [F] The Selene layer has 432 family A rows with n = 4, 6, 8 and 10 and 36 B1 rows from 12 circuits. [F] The proxy fit used 540 A rows with n up to 8 and 35 B1 rows from 7 circuits. [F] The target is the depolarised sampled F_H divided by the noiseless sampled F_H at the same seed. [F] On the 345 rows with a proxy value this ratio differs from the exact proxy F_H by 0.0022 in mean absolute value, and the raw sampled F_H differs by 0.0058. [O] The ratio is the closest available match to the exact fidelity that the proxy fit used.

[F] Under circuit-grouped cross-validation the refit matches the proxy fit, with pooled R^2 = 0.935 for OLS and 0.974 for GBR. [F] Training on A and testing on B1 gives R^2 = 0.859 for ridge and 0.848 for GBR. [F] The unfitted physics baseline drops on B1 from R^2 = 0.785 on the proxy to 0.216 on Selene. [F] Training on B1 and testing on A still fails, with R^2 = 0.327 for GBR and negative R^2 for all other models. [F] The largest OLS terms on A remain n_qubits, log2_support and n_2q_offsupport with t = 14.7, -14.9 and 15.9. [O] The Selene layer confirms the two proxy conclusions. Transfer runs from A to B1 only, and within a family the fitted models interpolate.

model	split	R^2 proxy	R^2 Selene	MAE(F_H) proxy	MAE(F_H) Selene
physics	A to B1	0.785	0.216	0.0151	0.0201
ols	A to B1	0.560	0.356	0.0252	0.0333

ridge	A to B1	0.840	0.859	0.0150	0.0154
gbr	A to B1	0.805	0.848	0.0083	0.0140
physics	B1 to A	-3.513	-3.760	0.0430	0.0464
ols	B1 to A	-261.894	-313.944	0.1971	0.6838
ridge	B1 to A	-0.605	-98.831	0.0259	0.2673
gbr	B1 to A	0.326	0.327	0.0184	0.0234
physics	cv5g pooled	-2.739	-2.773	0.0413	0.0444
ols	cv5g pooled	0.938	0.935	0.0051	0.0065
ridge	cv5g pooled	0.938	0.936	0.0052	0.0065
gbr	cv5g pooled	0.971	0.974	0.0017	0.0035

[F] Test rows: A to B1 35 proxy and 36 Selene, B1 to A 540 and 432, cv5g pooled 575 and 468. Source: `results/cost_model_selene.md`.

[F] The refit is reproduced by the three Selene cost-model rows of the script table in Section 5. `model/cost_model.py --prefix selene_` writes `results/cost_model_selene_metrics.csv` and `results/cost_model_selene_coefficients.csv` directly. A rerun on 8 October 2026 reproduced both files byte for byte.

5. Reproducibility

[F] Environment, confirmed with `pip show` on 7 October 2026 in the development container:

Package	Version
Python	3.13.16
pytket	2.18.4
pytket-quantinuum	0.59.3
pyzx (PyPI)	0.10.7
Quantinuum pyzx fork (family B1 sources only, `third_party/pyzx-fork/`)	commit e695d6d7
qiskit	2.5.2
pygridsynth	2.0.0
selene-sim	0.3.4
guppylang	1.1.2
hugr	0.18.8
numpy	2.5.3
pandas	2.3.3
scipy	1.18.1
scikit-learn	1.9.1
matplotlib	3.11.2
aqora	0.35.1
qnexus	0.51.1
marimo	0.25.1

[F] Seeds. `numpy.random.default_rng(20261006)` generates the family A couplings (`bench/gen_trotter.py`, `SEED`) and the family B2 Haar targets (`bench/gen_haar_b2.py`, `SEED`), with `seed_transpiler=0` in the qiskit transpile call of B2. Each generator draws from one stream in a fixed order, so the instance set does not depend on the order of writing. `noise/run_proxy.py` uses `default_rng(0)` only to pick the circuits of its parser self-check. `selene/run_selene.py` seeds every Selene run with `sha256("<pipeline>/<file>/<tag>")[:4]` and stores the value in the column `seed`. `model/cost_model.py` uses seed 0 for the fold assignment and `random_state=0` for the gradient-boosting model.

[F] Scripts, all run from the repository root. Inputs and outputs are the paths each script reads and writes.

Step	Command	Inputs	Outputs
Family A circuits	`python bench/gen_trotter.py`	none (seed 20261006)	`bench/family_A/*.qasm`, `*_meas.qasm`, `*.npy`, `manifest.csv`
Family B1 circuits	`python bench/build_family_B1.py`	`third_party/pyzx-fork/circuits/`	`bench/family_B1/*.qasm`, `manifest.csv`
Family B1x circuits	`python bench/build_family_B1x.py` (defaults `--min-qubits 13 --max-qubits 20 --max-gates 3000`)	`third_party/pyzx-fork/circuits/`	`bench/family_B1x/*.qasm`, `manifest.csv`
Family B2 circuits	`python bench/gen_haar_b2.py`	none (seed 20261006)	`bench/family_B2/*.qasm`, `*_target.npy`, `manifest.csv`
Family A compilation	`python pipelines/run_bench.py --workers 2`	`bench/family_A/`	`results/family_A_counts.csv`, `results/family_A_summary.csv`, `results/compiled/<pipeline>/*.qasm`
Family B1 compilation	`python pipelines/run_bench_B.py --manifest bench/family_B1/manifest.csv --prefix family_B1 --workers 2 --timeout 900`	`bench/family_B1/`	`results/family_B1_counts.csv`, `results/family_B1_summary.csv`, `results/compiled/<pipeline>/*.qasm`
hwb8 long run	`python pipelines/run_bench_B.py --manifest bench/family_B1/manifest.csv --files hwb8 --pipelines P1,P2 --prefix family_B1_hwb8_long --timeout 5400 --workers 2`	`bench/family_B1/hwb8.qasm`	`results/family_B1_hwb8_long_counts.csv`, `results/family_B1_hwb8_long_summary.csv`, `results/family_B1_hwb8_long_run.log`
Family B1x compilation, count-only	`python pipelines/run_bench_B.py --manifest bench/family_B1x/manifest.csv --prefix family_B1x --pipelines P0ref,P0,P2 --no-check`	`bench/family_B1x/`	`results/family_B1x_counts.csv`, `results/family_B1x_summary.csv`, `results/family_B1x_run.log`, `results/compiled/{P0ref,P0,P2}/<file>.qasm`
Family B2 compilation	`python pipelines/run_b2.py --workers 2`	`bench/family_B2/`	`results/family_B2_counts.csv`
Noise proxy, family A	`python noise/run_proxy.py --noise h22e --family A --workers 2`	`results/compiled/{P0ref,P0,P1,P2,P3}/`, `bench/family_A/*.npy`	`results/family_A_proxy_h22e.csv`

Noise proxy, family B1	`python noise/run_proxy.py --noise h22e --family B1 --workers 2`	`results/compiled/{P0ref,P0,P1,P2,P3}/`, `bench/family_B1/manifest.csv`	`results/family_B1_proxy_h22e.csv`
Proxy figure	`python noise/run_proxy.py --figure-h22e`	the two proxy CSVs	`results/fig_proxy_h22e.{pdf,png}`, `results/proxy_h22e_fits.log`
Certificates	`python barrier/run_certificates.py`	`bench/family_B2/*.qasm`, `bench/family_B1/{barenco_tof_3,mod5_4,qft_4,tof_3}.qasm`, `results/family_B2_counts.csv`, `results/family_B1_counts.csv`	`results/certificates.csv`, `results/certificates.md`
Selene runs	`HTTPS_PROXY= https_proxy= HTTP_PROXY= http_proxy= NO_PROXY= no_proxy= python selene/run_selene.py --mode both --procs 2 --shots 2000`	`results/compiled/{P0,P1,P2}/`, `bench/family_A/*.npy`, `bench/family_B1/manifest.csv`, the two proxy CSVs	`results/selene_noiseless.csv`, `results/selene_depolarizing.csv`, `results/selene_run.log`
Selene summary	`python selene/make_summary.py`	the two Selene CSVs, `results/selene_calibration.txt`, `results/selene_recheck.txt`	`results/selene.md`
Cost model	`python model/cost_model.py`	`results/family_A_proxy_h22e.csv`, `results/family_B1_proxy_h22e.csv`	`results/cost_model_metrics.csv`, `results/cost_model_coefficients.csv`, `results/cost_model.md`, `results/fig_cost_model.{pdf,png}`
Selene cost-model inputs	`python model/build_selene_cost_inputs.py`	`results/selene_depolarizing_full.csv`, `results/family_A_counts.csv`, `results/family_B1_counts.csv`	`results/cost_model_selene_input_A.csv`, `results/cost_model_selene_input_B1.csv`
Selene cost-model refit	`python model/cost_model.py --family-a results/cost_model_selene_input_A.csv --family-b results/cost_model_selene_input_B1.csv --prefix selene_`	the two Selene input CSVs	`results/cost_model_selene_metrics.csv`, `results/cost_model_selene_coefficients.csv`, `results/cost_model_selene_fig.{pdf,png}`, `results/cost_model_selene_autoreport.md`
Selene cost-model summary	`python model/report_selene_cost.py`	the Selene and proxy metric and coefficient CSVs, the two Selene input CSVs, `results/selene_depolarizing_full.csv`	`results/cost_model_selene.md`
Published T-counts of Table C2	`python report/build_literature_tcounts.py results/literature_tcounts.csv`	`results/family_B1_counts.csv`, `results/family_B1x_counts.csv`, values copied from the six papers of Section 3.4	`results/literature_tcounts.csv`
Tables and figures	`python report/make_tables.py`	`results/*.csv`	`report/tables/*.md`, `report/figures/*.png`, injected into the report
Emulator jobs	`python3 submit/syntax_check.py` and `python3 submit/run_emulator.py`, with platform, shots and options of each job recorded in `results/emulator_jobs.csv`	`results/compiled/<pipeline>/`	`results/emulator_jobs.csv`, `results/emulator_nexus_H2-Emulator.csv`, `results/emulator/<platform>/<pipeline>/<file>.json`

[F] The Selene command clears every proxy variable because Selene streams its results over a local TCP socket that the container proxy intercepts (`results/selene.md`, section 1).

[F] Tests. Each file was run once with the versions above, on 7 October 2026 unless the table gives 8 October. Six files run as scripts. Six files run under pytest. These are `selene/test_selene.py`, `submit/test_submit.py`, `submit/test_h22e_stats.py`, `submit/test_h22e_pooled.py`, `value/test_value_ledger.py` and `value/test_value_by_application.py`.

File	Command	Result	Wall time
`bench/test_gen_trotter.py`	`python bench/test_gen_trotter.py`	12 of 12 passed	58 s
`noise/test_dm_sim.py`	`python noise/test_dm_sim.py`	17 of 17 passed	3 s
`barrier/test_certificate.py`	`python barrier/test_certificate.py`	69 of 69 checks passed	29 s (44 s in `results/verification_2026-10-07.md`)
`pipelines/test_compile.py`	`python pipelines/test_compile.py`	6 of 6 passed	2 s
`selene/test_selene.py`	proxy variables cleared, `python -m pytest selene/test_selene.py -q`	6 of 6 passed (5 functions, one parametrised twice)	7 s
`model/test_cost_model.py`	`python model/test_cost_model.py`	7 of 7 passed, rerun 8 October after the `--prefix` change	under 1 s
`submit/test_submit.py`	`python -m pytest submit/test_submit.py -q`	11 of 11 passed, 8 October, offline with no credentials	4 s
`submit/test_h22e_stats.py`	`python -m pytest submit/test_h22e_stats.py -q`	7 of 7 passed, 8 October	4.2 s
`submit/test_h22e_pooled.py`	`python -m pytest submit/test_h22e_pooled.py -q`	9 of 9 passed, 8 October	6.2 s
`value/test_value_ledger.py`	`python -m pytest value/test_value_ledger.py -q`	5 of 5 passed, 8 October	2.9 s
`value/test_value_by_application.py`	`python -m pytest value/test_value_by_application.py -q`	5 of 5 passed, 8 October	1.1 s
`report/test_c2_literature.py`	`python report/test_c2_literature.py`	all checks passed (12 assert statements), 8 October	0.6 s

[F] The compile tests check every pipeline against the unitary of the input and check that the measured QASM encodes the implicit qubit permutation left by `allow_swaps=True`. The certificate tests reproduce the finite verifications of the barrier manuscript (the counts 24, 1536 and 18384). The Selene tests check the bit order of the returned array with a permuted measure map and check that a seed reproduces the counts. The cost-model tests check that OLS recovers a known law without noise and that the physics baseline is exact on its own law. The submit tests run offline. They check the native gate set, the HQC formula, the bit order of returned counts, F_H and TVD, the job ledger, the job plans, the credential detection and the rubric rendering.

[F] Checks recorded per row in every counts CSV: `check_kind` and `check_dev`, `dist_maxdiff` and `qasm_dist_maxdiff`, `check_ok` and `status`, `pyzx_angle_err`, and the timings `t_zx`, `t_tket` and `t_total`. [F] The flag `--no-check` of `pipelines/run_bench_B.py` skips the unitary, statevector and distribution checks and keeps the gate-set check. Under it `check_kind` is `none`, the check columns are NaN and `status` is `ok-unchecked` for a native circuit and `FLAGGED-unchecked` otherwise. Family B1x is the only family compiled under it, and all 24 of its rows are `ok-unchec`

ked with a total wall time of 36.8 s.

[F] Timeouts. pipelines/run_bench.py and pipelines/run_bench_B.py default to 600 s per (circuit, pipeline) and pipelines/run_b2.py to 900 s. The family B1 run used 900 s, under which hwb8 timed out under P1 and P2, so the first results/family_B1_counts.csv carried status = timeout on those two rows (103 of 105 rows ok). A rerun with --timeout 5400 on P1 and P2 only started on 7 October at 21:37 and completed both. P1 finished in 885.6 s wall with 8305 ZZPhase gates and P2 in 843.8 s wall with 5587 ZZPhase gates. Both rows have status = ok with check_dev at most 1.7e-13 and a T-count after the ZX stage of 3517 (results/family_B1_hwb8_long_run.log, results/family_B1_hwb8_long_counts.csv). [F] The two rows replaced the timeout rows in results/family_B1_counts.csv, results/family_B1_summary.csv was regenerated with summarise from pipelines/run_bench_B.py and python report/make_tables.py was rerun, so every table built from the main CSV shows hwb8 under all seven pipelines and all 105 rows are ok. The file before the merge is kept as results/family_B1_counts_before_hwb8_merge.csv. [O] Both wall times lie under the earlier 900 s limit, so the earlier timeouts came from two workers sharing the two cores with the other 13 circuits, not from the circuit itself.

[F] Verification. Seven verification reports by agents that did not produce the results are in results/verification_2026-10-0*.md. results/verification_2026-10-07.md recomputed the seven claims of claude/status-2026-10-06.md from the raw CSVs and QASM files. Six were confirmed and one mismatched: the status document said P2 beats P0 on 13 of 14 completed B1 circuits and the recomputation gives 12 of 14 with two ties. One figure was unverifiable, a 1064-gate SK word length that depends on an unstated approximation depth. results/verification_2026-10-07b.md recomputed six further claims (family A files and manifest, family_A_counts.csv, the H2-2E proxy CSVs and their no-intercept fits, four proxy rows rerun, certificates.csv against a direct recomputation, ten report table cells): all confirmed, none mismatched. That report also diagnosed the crash of the first hwb8 rerun, in which --files hwb8 matched against hwb8.qasm selected zero jobs, and fixed pipelines/run_bench_B.py before the 5400 s run above was started.

- [F] results/verification_2026-10-07c.md rechecked report v2 item by item, from Sections 1 and 2 to the prose rules.
- [F] results/verification_2026-10-08a.md rechecked the family B1x numbers of report v3 and recompiled tof_10 and rc_adder_6.
- [F] results/verification_2026-10-08b.md checked Section 1, Section 3.7.1, the Section 6 hardware sentence, the scoring map and the v5 changelog.
- [F] results/verification_2026-10-08c.md checked Table C2 and the overclaim fixes.
- [F] results/verification_2026-10-08d.md checked the report from the title to the References. Its fixes F2 to F31 are applied in this draft. F32 is not applied because the generated captions it names are shared with drafts v1 to v4.

6. Limitations

[F] ZX extraction worsens the ZZPhase count. On family A the pipelines P1, P3 and P4 are above P0 on 180 of 180 circuits, with ZZ ratios from 1.28 to 10.32 (P1), 1.31 to 7.91 (P3) and 1.44 to 10.36 (P4). On family B1 (results/family_B1_counts.csv) P1 is above P0 on 14 of 15, P3 on 14 of 15 and P4 on 15 of 15. Only teleport reduction, which does not extract, helps on H2.

[F] On the Trotter family no pipeline goes below P0 in ZZPhase count. Under P0 alone the ZZPhase count equals one gate per two-body term on 180 of 180 circuits, where n_2q equals n_cx / 2 of the manifest. No pipeline has fewer ZZPhase gates than P0 on any family A circuit, 0 of 180 for each of P0ref, P0L1, P1, P2, P3 and P4. The family measures the cost of a rewrite instead of its benefit. [O] The Trotter family is the right control, because a Trotter circuit is what an H2 user runs most, and a method that damages it has to be caught.

[F] A single-qubit T gate is an Rz on H2, which is free in HQC and absent from the gate-noise model. The T-count layer of the ledger has no effect on any H2 number in this report. Its value is for a fault-tolerant target.

[F] The noise proxy is a density-matrix simulation with the published H2-2E default parameters (p1 = 7.3e-5, p2 = 1.29e-3, p_meas = (9e-4, 1.8e-3), p_init = 4e-5) and depolarising gate channels only. The published H2-2E model has in addition a spontaneous-emission branch that applies leakage half of the time, measurement and initialisation crosstalk, quadratic and linear dephasing during transport and idling, and a two-qubit fault probability that scales with the RZZ angle (docs/platform_notes.md, section 3.2). The proxy has none of these. The Selene cross-check in results/selene.md compares two depolarising models that share the same per-gate convention, so the agreement of 32 of 34 runs within two standard deviations for the like-for-like variant confirms the implementation of the proxy and its measure-map handling, not its fidelity to H2-2E. [F] In the compiled P0 and P2 circuits of the 14 B1 circuits other than hwb8, 1421 of the 1661 ZZPhase gates have angle pi/2, where the scaling gives p2. Another 238 have angle pi/4, where it gives 0.62 p2. Two have angle 0.375 pi. [O] The angles of family A are small, so the published scaling lowers the two-qubit fault probability below p2 and the proxy is pessimistic there.

[F] Nothing in this report has run on hardware. The 22 H2-Emulator rows of Section 3.7 and the 18 H2-2E jobs of Section 3.7.1 are emulator runs. vbe_adder_3 at 4000 shots per pipeline is the only emulator comparison that resolves P2 against P0 in fidelity, with z = +2.38. No other emulator comparison of P2 against P0 is resolved. The 468 noiseless, the 68 depolarising and the 468 full depolarising Selene runs are local. They use the Quest statevector backend and do not contain the H2-2E noise model. The submission cell of report/notebook_skeleton.py has not been executed. Its stored marimo session report/__marimo__/session/notebook_skeleton.py.json stops at the Submit to QPU gate. The platform jobs were submitted by submit/syntax_check.py and submit/run_emulator.py.

[F] The cost model is fitted on proxy fidelities, not on emulator output. Under gradient boosting with the grouped split cv5g held-out whole circuit classes give R^2 = 0.990 on family A rows and 0.830 on family B1 rows. This measures interpolation within the families. Across families the evidence runs in one direction. Trained on family A the boosted model predicts the 7 family B1 circuits with R^2 = 0.805 and MAE(F_H) 0.0083. Ridge reaches R^2 = 0.840 and the unfitted physics baseline 0.785. Trained on the 35 family B1 rows no model predicts family A and the best R^2 is 0.326. [O] Seven B1 circuits do not establish generalisation, and learned rewriting itself is not attempted in this submission.

[F] The certificate bounds the T-count of exact rewrites and is computed on the channel of size 4^n x 4^n. It is applied to family B2 at n = 2 and n = 3 and to four family B1 circuits at n = 5. On family B2 the floor is at least 99.28 % of the post-ZX T-count on every instance. On the four B1 circuits the floor is 25.0 % (barenco_tof_3), 25.0 % (mod5_4), 55.2 % (qft_4) and 26.7 % (tof_3) of the post-ZX T-count, so there the certificate leaves most of the remaining T-count unaccounted for. [O] The B1 floors are loose because those circuits are hand-written with many Clifford collapses, not synthesised words, and the manuscript's near-rigidity result covers synthesised words only. An approximate resynthesis to a new precision is outside the scope of the certificate.

[F] hwb8 under P1 and P2 completed only in the 5400 s run of Section 5 (P1 8305, P2 5587 ZZPhase gates), and those two rows are

the ones in `results/family_B1_counts.csv` and Table B1. P2 is below P0 on hwb8 by 5.50 % (5912 to 5587) and P2 is below P0 on 13 of 15 family B1 circuits with two ties. `[F]` The 8 circuits of family B1x (Section 3.2.1) were compiled with `--no-check` and have no unitary, statevector or distribution check. Their counts are not evidence of exactness and are kept apart from the family B1 totals.

`[F]` The implementation of the per-step bookkeeping of Lemma mqval(b) of the barrier manuscript found that the "reduction fires exactly once" half of that lemma fails on the word T T (`barrier/NOTES.md`, section 8), and the floor of Corollary cert does not use that half.

`[F]` Table B2 has five circuits per row against 50 and 8 in the manuscript. The agreement of the means is a consistency check, not a statistical result.

7. Team and roles

`[F]` The team members are C.-F. Kam, A. Mahasinghe, D. De Silva, Karunarathne, F. Cadet, Bessafi and K. De Silva.

`[F]` C.-F. Kam is the first author of the exactness-barrier manuscript arXiv:2608.22801 (Reference 1). `[O]` Kam is the team owner on Aqora as stated by the team. No artifact in the repository records it.

`[O]` Roles of the other members to be confirmed by Kam before submission.

References

1. C.-F. Kam et al., An Exactness Barrier for ZX-Calculus Optimization of Synthesized Clifford+T Circuits, arXiv:2608.22801. The TeX source is not distributed. `[F]` arXiv:2608.22801 resolves to this title with Kam as first author (`results/verification_2026-10-08b.md`, claim 15).
2. D. De Silva et al., Numerical Evaluation of ZX Calculus Optimization for Solovay Kitaev Quantum Circuit Synthesis, arXiv:2608.22810. The TeX source is not distributed. `[F]` arXiv:2608.22810 resolves to this title with D. De Silva as first author, fetched 8 October 2026.
3. A. Kissinger and J. van de Wetering, Reducing the number of non-Clifford gates in quantum circuits, Phys. Rev. A 102, 022406 (2020), arXiv:1903.10477. Table 1 checked on 7 October 2026 (results/literature_tcount_check.md).
4. Quantinuum, H2 Product Data Sheet, version 4.00, 17 September 2025, https://docs.quantinuum.com/systems/_static/assets/data_sheets/Quantinuum%20H2%20Product%20Data%20Sheet.pdf.
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7. Quantinuum, Singapore Grand Challenge code samples, https://github.com/Quantinuum/sg_grand_challenge.
8. Quantinuum, Selene emulation framework, https://docs.quantinuum.com/selene/ and https://github.com/quantinuum/selene.
9. pytket-quantinuum 0.59.3, `QuantinuumBackend.default_compilation_pass`, reproduced in `pipelines/compile.py:quantinuum_default_pass` with line references.
10. PyZX 0.10.7 and the Quantinuum pyzx fork, commit e695d6d7, for the family B1 sources.